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Netrin signaling mediates survival of dormant epithelial ovarian cancer cells

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Abstract

Dormancy in cancer is a clinical state in which residual disease remains undetectable for a prolonged duration. At a cellular level, rare cancer cells cease proliferation and survive chemotherapy and disseminate disease. We created a suspension culture model of high grade serous ovarian cancer (HGSOC) dormancy and devised a novel CRISPR screening approach to identify survival genes in this context. In combination with RNA-seq, we discovered the Netrin signaling pathway as critical to dormant HGSOC cell survival. We demonstrate that Netrin-1, -3, and its receptors are essential for low level ERK activation to promote survival, and that Netrin activation of ERK is unable to induce proliferation. Deletion of all UNC5 family receptors blocks Netrin signaling in HGSOC cells and compromises viability during the dormancy step of dissemination in xenograft assays. Furthermore, we demonstrate that Netrin-1 and -3 overexpression in HGSOC correlates with poor outcome. Specifically, our experiments reveal that Netrin overexpression elevates cell survival in dormant culture conditions and contributes to greater spread of disease in a xenograft model of abdominal dissemination. This study highlights Netrin signaling as a key mediator HGSOC cancer cell dormancy and metastasis.

eLife assessment

The authors further corroborated their model that Netrin signaling promotes survival and dissemination of non-proliferating ovarian cancer cells. These **valuable** results were found to be of significant potential interest to cancer biologists in as much as they address gaps in knowledge pertinent to the mechanisms underpinning ovarian cancer spread. In general, it was thought that **solid** experimental evidence was provided to support the role of Netrin signaling in fueling ovarian cancer progression.

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Introduction

Relapse from initially effective cancer treatment represents one of the greatest barriers to improving outcomes (1). Dormant or minimal residual disease that is not clinically detectable is a common source of relapse, particularly in a metastatic setting (2). This has motivated the search for, and study of, disseminated cancer cells. Cellular dormancy has been characterized in cancer cells derived from a variety of primary disease sites including breast, prostate, melanoma, and others (3, 4). Disseminated cells often occupy specific cellular niches in perivascular space or in bone marrow (5). They adopt catabolic metabolism (2), and withdraw from the cell cycle (4). Stem like phenotypes and epithelial to mesenchymal transition also characterize the biology of these cells as they survive dissemination and lay quiescent in distant tissues (1, 3, 5). Across a variety of disease sites this dormant state is characterized by diminished Ras-MAPK signaling and elevated p38 activity (3, 6). To develop treatments specifically tailored for dormant cells, a systematic search for vulnerabilities unique to dormancy is needed. To this end, high grade serous ovarian cancer (HGSOC) disseminates in ascites as growth arrested cellular aggregates (7). This has been modeled in culture (8), and studies confirm active induction of cell cycle arrest mechanisms (9, 10), induction of autophagy (11, 12), and other known characteristics of dormancy (13). This suggests that HGSOC spheroids are ideal to search for survival dependencies specific to dormant cells.

High grade serous ovarian cancer (HGSOC) is the deadliest gynecologic malignancy, accounting for more than 70% of cases (14). Its poor prognosis is largely attributed to late-stage diagnosis with metastases already present. Metastatic spread is facilitated by multicellular clusters of tumor cells, called spheroids, that are shed into the peritoneal cavity and disseminate to nearby organs (15). The non-proliferative state of spheroid cells renders them chemoresistant and contributes to disease recurrence, emphasizing the need to understand the biology of these cells (16). Dormant HGSOC cells are characterized by similar metabolic changes, EMT, and cytokine signaling as dormant cancer cells from other disease sites (3, 13). Mechanisms that mediate their dormant properties include AMPK-LKB1 signaling that induces autophagy (17, 18), STAT3 and FAK for survival and chemoresistance (19, 20), and DYRK1A signaling to arrest proliferation (10). Unfortunately, therapeutic approaches to target these pathways in ovarian cancer have yet to emerge, indicating that more candidates are needed.

Netrins are a family of secreted factors that reside in the extracellular matrix and were first identified in axon guidance in the developing nervous system (21, 22). Together with their receptors, Netrins have since been implicated in cancer progression (23). Netrin-1 is the most studied and its misregulation in cancer leads to pro-survival signals (24). Netrin interactions with DCC and UNC5 homologs (UNC5H, UNC5A-D) convert pro-death signals by these dependence receptors to survival signals (25). Netrin signals through PI3K-AKT and ERK-MAPK are known to participate in orientation decisions in development by providing guidance cues (26–28), while loss of ligand binding leads DCC or UNC5H to stimulate dephosphorylation of DAPK1 that sends pro-death signals (29, 30). The gene encoding DCC is frequently deleted in cancer (24), and Netrin-1 blocking antibodies are in Phase II clinical trials for endometrial and cervical carcinomas to inhibit metastatic growth (25). Alternative receptors for Netrins such as NEO, DSCAM, and Integrins also mediate their signals (23), creating a myriad of possible pathways for Netrins to regulate cancer cells. Lastly, functional roles for Netrin-3, -4, and -5 are still relatively unexplored compared to Netrin-1 (31), although their expression patterns suggest roles in advanced forms of some cancers (32). A functional role for Netrins and their receptors in dormancy, or other aspects of HGSOC spheroid biology have yet to be reported.

In this study we utilized a suspension culture system to interrogate survival mechanisms in dormancy using HGSOC spheroids. To maximize discovery of gene loss events that compromise viability in dormancy, we devised a novel genome-wide CRISPR screen approach that we term 'GO-CRISPR'. In addition to the standard CRISPR workflow, GO-CRISPR utilizes a parallel screen in which 'guide-only' cells lacking Cas9 are used to control for stochastic changes in gRNA abundance in non-proliferative cells. This screen identified multiple Netrin ligands, DCC, UNC5Hs, and downstream MAPK signaling components as supporting survival of HGSOC cells in dormant culture conditions. A transcriptomic analysis of HGSOC spheroids also revealed Netrin signaling components were enriched in HGSOC spheroids compared to adherent cells, and that in the absence of the DYRK1A survival kinase Netrin transcriptional increases were lost. We show that multiple Netrins, UNC5 receptors, and the downstream MEK-ERK axis is crucial for survival signaling in dormant cell culture. Importantly, Netrin activation of ERK is restricted to spheroid cultures and is unable to stimulate proliferation, consistent with its role in dormant survival. We blocked Netrin signaling by deleting UNC5 receptors and using xenograft approaches demonstrated that loss of Netrin signaling compromises survival of HGSOC cells, specifically in the dormant stage of dissemination. Furthermore, we show that overexpression of Netrin-1 and -3 are associated with poor prognosis in HGSOC and their overexpression in culture increases survival in suspension culture. Netrin overexpression in xenografts contributes to increased disease dissemination in a model of metastasis. Our study highlights Netrin-ERK signaling as a requirement for spheroid survival and suggests it may be a potential therapeutic target for eradicating dormant HGSOC cells.

Results

Axon guidance genes are essential factors for HGSOC spheroid viability

To dissect genetic dependencies of dormant ovarian cancer cells, we utilized a suspension culture system that encourages cellular aggregation, leading to non-proliferative spheroids that resemble dormant aggregates found in ovarian cancer patient ascites (7, 10, 13, 33). **Fig. 1A** illustrates withdrawal from cellular proliferation by HGSOC cells in suspension as evidenced by reduced expression of Ki67 and increased expression of the quiescence marker p130 (34). Furthermore, phospho-western blots for ERK demonstrate reduced but not absent phosphorylation, and phospho-p38 either remains consistent or is increased. Reduced phospho-ERK and elevated phospho-p38 are indicative of the characteristic signaling switch reported to underlie most paradigms of cancer cell dormancy (5). These findings are relatively consistent among the three HGSOC cell lines; iOvCa147, OVCAR8, TOV1946 selected for use in our screen. To discover genes required for survival in spheroids, we utilized a new approach called GO-CRISPR (Guide Only CRISPR) that we developed specifically for survival gene discovery in three-dimensional spheroid cell culture conditions (**Fig. 1B**). GO-CRISPR incorporates sgRNA abundances from non-Cas9-expressing cells to control for stochastic (non-Cas9 dependent) effects that are a challenge in three-dimensional culture conditions with slow or arrested proliferation (35). We used sequencing to determine sgRNA identity and abundance in the library, initially infected cells, and in cells following 48 hours of suspension culture (**Fig. 1B**). Read count data was analyzed using a software package that we developed to analyze GO-CRISPR data (summarized in **Fig. 1 - Fig. Sup. 1A&B**). We selected genes that had a Spheroid/Adherent Enrichment Ratio (ER) less than 1 (P<0.05) because some genes become depleted in adherent growth before being transferred to suspension culture conditions and this ratio selects for genes with a stronger effect in suspension when cells are not proliferating (**Fig. 1B**). To assess reliability of screen data, we investigated the classification of non-targeting control sgRNAs. We computed a receiver operator characteristic curve (ROC) and determined the area under the curve for each cell line (**Fig. 1 - Fig. Sup. 1C**). Greater than 95% of sgRNAs in iOVCA147 and OVCAR8 screens were classified as non-essential, while 83% of sgRNAs in TOV1946 cells were non-essential. In addition, we compared GO-CRISPR screen ER values with viability data from siRNA knock down for 35 individual genes we have previously studied in spheroid survival assays in OVCAR8 and iOVCA147 cells (**Fig. 1 - Fig. Sup. 1D**)(10). This revealed a statistically significant correlation between the two sets of data, suggesting that genes we previously knew supported spheroid viability (or were dispensable) were classified similarly by our screen. These controls suggested our screen data was reliable for identifying new survival dependencies.

We found 6,717 genes with an ER below 1 in iOvCa147 cells; and 7,637 genes in TOV1946; and 7,640 genes in OVCAR8 cells. Among these genes; 1,382 were common to all three cell lines (**Fig. 1D**). We performed pathway enrichment analysis on these common genes and identified several significantly enriched pathways. These include primary and secondary Reactome gene sets such as signal transduction, metabolism, and signaling by GPCR (**Fig. 1E**). In addition, we identified a tertiary category called 'axon guidance,' that includes numerous components of the Netrin signaling pathway that scored highly.

Expression of Netrin ligands and receptors are enriched in HGSOC spheroids

DYRK1A was previously discovered to support survival of ovarian cancer cells in suspension culture (10). Since DYRK1A phosphorylates RNA polymerase II in transcription initiation (36), we sought to determine if it regulates gene expression in spheroid survival. We created *DYRK1A*$^{-/-}$ iOvCa147 cells and confirmed their deficiency for survival in suspension (**Fig. 2 - Fig. Sup. 1A-E**). Using iOvCa147 and *DYRK1A*$^{-/-}$ derivatives, we performed RNA-seq on adherent and spheroid cells. We collected spheroids following a 6-hour incubation in suspension because overexpression of known cell cycle transcriptional targets due to DYRK1A loss were first evident at this time point, but loss of viability due to DYRK1A deficiency had not yet occurred (10). We compared iOvCa147 adherent cells to 6-hour suspension culture to identify transcriptional changes that occur as these cells transitioned from adherent conditions to suspension conditions (**Fig. 2A**). We identified 1,834 genes that were upregulated, and pathway enrichment analysis identified many of the same categories as our CRISPR screen including axon guidance (**Fig. 2B**). We then compared iOvCa147 spheroid cells to *DYRK1A*$^{-/-}$ spheroid cells to identify mRNA expression changes that were lost through DYRK1A deficiency (**Fig. 2C**). We identified 744 genes that were downregulated and these also belonged to many of the same pathways, including axon guidance (**Fig. 2D**).

We compared enriched pathways from these two RNA-seq analyses with those that were identified in our GO-CRISPR screens to identify gene expression programs in spheroids whose products are essential for cell survival. 78 Reactome pathways were commonly enriched across the GO-CRISPR screens and transcriptomic analyses, including 78 of the 83 pathways identified in RNA-seq from *DYRK1A*$^{-/-}$ spheroid cells (**Fig. 2E**). The axon guidance pathway was one of the most enriched among these common pathways (**Fig. 2F**). These data reveal that our GO-CRISPR screen and transcriptomic analyses converge on this highly specific tertiary Reactome category of axon

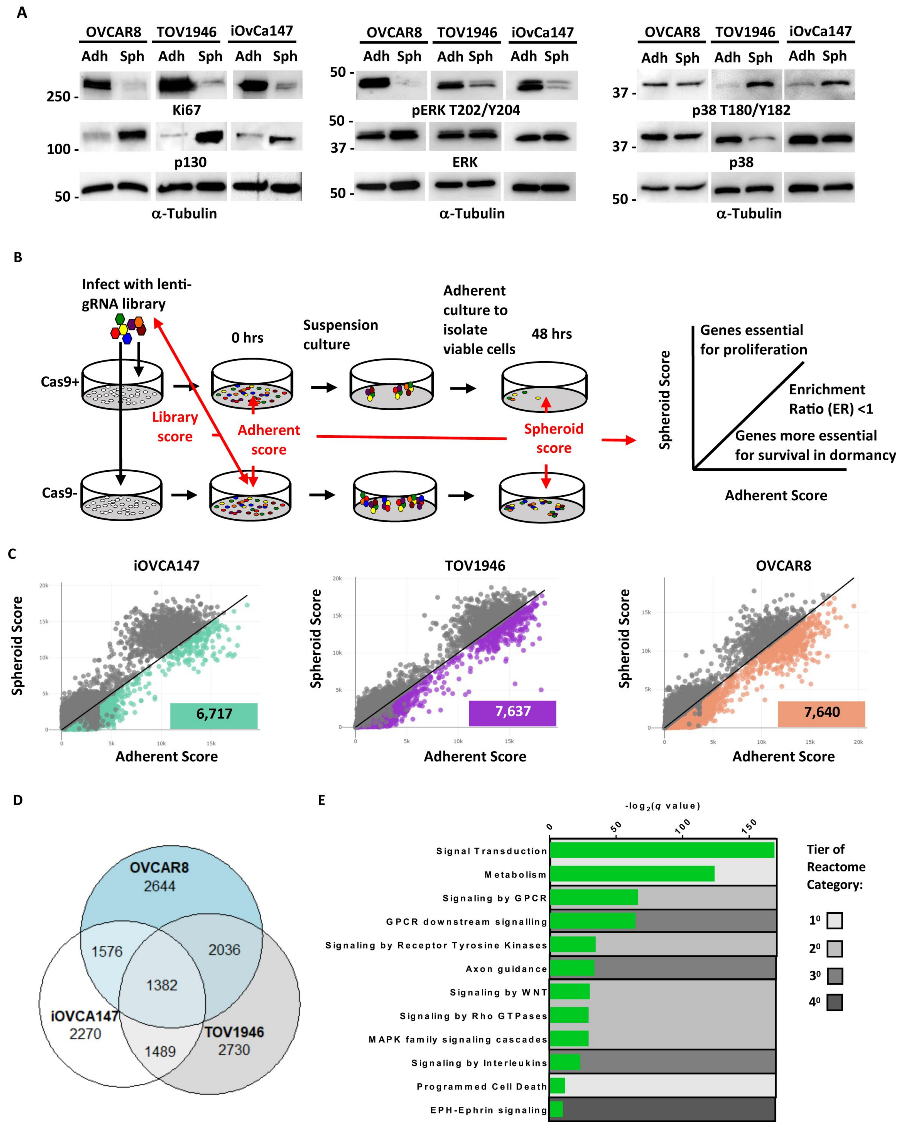

Figure 1.

GO-CRISPR screens implicate axon guidance pathways as supporting HGSOC spheroid cell viability.

A) iOvCa147, TOV1946, or OVCAR8 cells were cultured under adherent conditions (Adh) or in suspension to induce spheroid formation (Sph). Lysates were prepared and analyzed by western blotting for the proliferation marker Ki67 and the quiescence marker p130, and phosphorylated and total levels of ERK and p38. Tubulin was blotted as a loading control. **B)** Flow chart of GO-CRISPR screening used for each of iOvCa147, TOV1946, or OVCAR8 control cells and derivatives of each that express Cas9. Cas9-positive cells (top row) and Cas9-negative cells (bottom row) were transduced with the GeCKO v2 pooled sgRNA library. After antibiotic selection, cells were expanded under adherent culture conditions (0 hours) before being transferred to suspension culture conditions to induce spheroid formation and select for cell survival. After 48 hours spheroids were transferred to standard plasticware to isolate viable cells. Red arrows indicate the relevant comparisons of sgRNA sequence abundance that were made to analyze screen outcomes. Genes with relatively greater effect on viability in suspension were selected by comparing their scores between adherent and suspension conditions and considering genes with an enrichment ratio (ER) of <1. **C)** Scatter plots representing the spheroid score on y-axis and adherent score on the x-axis calculated by TRACS for each gene in each cell line (iOvCa147, TOV1946, OVCAR8). Colored data points represent genes with ER < 1 and $p_{adj} < 0.05$ (paired t-test). **D)** Venn diagram illustrating overlap of genes identified as supporting cell viability in suspension culture from iOvCa147, TOV1946, and OVCAR8 cells. **E)** Graph depicting enriched pathways from ConsensusPathDB using the 1382 commonly identified genes from D. Categories are ranked by q-value. Tiers of Reactome categories are indicated by shading.

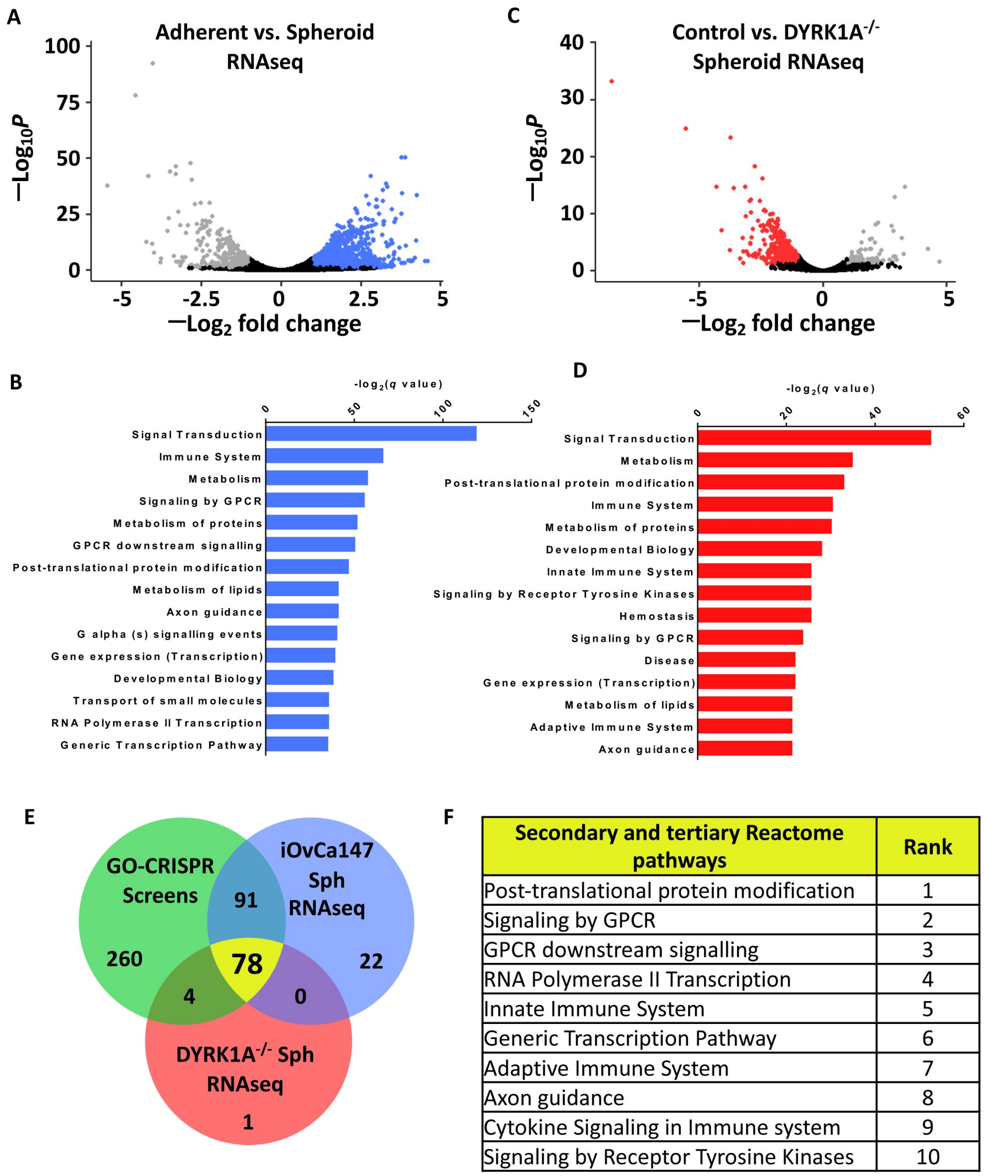

Figure 2.

Axon guidance pathway components are upregulated in iOvCa147 spheroid cells in a DYRK1A dependent manner.

A) RNA was isolated from iOvCa147 cells following culture under adherent conditions or in suspension conditions to induce spheroid formation for 6 hours. Triplicate independent cultures were processed for RNA-seq. A volcano plot shows differentially expressed genes in spheroid cells compared to adherent. 1,937 genes were found to be downregulated in iOvCa147 spheroid cells (log_2 fold change < 1, p_{adj} < 0.05 (Wald test), FDR 10%, highlighted in grey) and 1,834 genes were upregulated (log_2 fold change > 1, p_{adj} < 0.05, FDR 10%, highlighted in blue). **B)** Top 15 most significantly enriched pathways (p_{adj} < 0.05) whose genes were upregulated in suspension culture compared to adherent in RNA-seq analysis. A volcano plot showing differentially expressed genes in *DYRK1A*$^{-/-}$ spheroid cells compared to iOvCa147 spheroid cells. 744 genes were found to be downregulated in *DYRK1A*$^{-/-}$ spheroid cells (log_2 fold change < 1, p_{adj} < 0.05 (Wald test), FDR 10%, highlighted in red) and 96 genes were upregulated (log_2 fold change > 1, p_{adj} < 0.05 (Wald test), FDR 10%, highlighted in grey). **C)** Top 15 most significantly enriched pathways that were represented by downregulated genes in *DYRK1A*$^{-/-}$ suspension culture compared to control cells in suspension. **E)** Venn diagram depicting overlapping enriched pathways identified in GO-CRISPR screens in green; enriched pathways identified in upregulated genes in parental iOvCa147 spheroid cells in blue; and enriched pathways identified in downregulated genes in *DYRK1A*$^{-/-}$ spheroid cells in red. 78 pathways were commonly enriched in all three datasets (shown in yellow). **F)** Top 10 most significantly enriched pathways among the 78 identified in C.

guidance alongside much broader primary or secondary gene sets. This category includes Netrin ligands, receptors, and downstream signaling components, suggesting a previously unappreciated role in HGSOC spheroid viability.

We separately sought to confirm elevated expression of Netrins and some of their signaling components in HGSOC spheroid cell culture. We examined expression levels of ligands and receptors in the three cell lines used in our GO-CRISPR screen using RT-qPCR (**Fig. 3A-C**). The precise identities of upregulated genes varies between cell lines, however each cell line increases expression of some Netrins, at least one UNC5, and at least one of DCC, DSCAM, and Neogenin. This suggests that the Netrin pathway components are expressed in HGSOC cells may play a previously unappreciated role in dormancy and disease progression.

Since Netrin signaling has not been previously associated with dormancy we examined expression of Netrin-1 and -3 ligands in spheroids isolated directly from patient ascites by immunohistochemistry. First, we assessed proliferative markers in HGSOC tumor tissue and spheroids by staining for Ki67 and p130. As expected, solid HGSOC tumors display uniform Ki67 positivity and relative absence of p130 (**Fig. 4A-B**). Conversely, serial sections of spheroids demonstrated rare Ki67 nuclear staining and abundant and uniform p130 staining. This indicates that cells in these patient derived spheroids are quiescent (**Fig. 4C-E**). To investigate the presence of Netrins, we initially tested Netrin-1 and -3 antibody specificity using spheroids generated from OVCAR8 cells bearing knock out or over expression of Netrins (**Fig. 4** - **Fig. Sup. 1A-F**). Dormant patient derived spheroids were stained for Netrins and epithelial markers Cytokeratin and EpCAM. Since HGSOC is characterized by point mutations in p53 and its accumulation in the nucleus, sections were also stained for p53 to confirm cancer cell identity within these spheroids by high level nuclear staining (**Fig. 4F-H**). Overall, this confirms Netrin expression occurs in dormant HGSOC patient spheroids, precisely the context our screen indicates that they are critical for cell survival.

Netrin signaling stimulates ERK to support HGSOC spheroid survival

Netrin ligands and their dependence receptors are best known for their roles in axon guidance (37), but are appreciated to regulate apoptosis in cancer (23). **Fig. 5A** illustrates components of Netrin signaling and the frequency with which they were discovered to have an ER of <1 in our screens. In addition, intracellular signaling molecules known to be downstream of Netrin receptors that are shared with other enriched pathways identified in our screens, are also illustrated. To confirm screen findings, we independently knocked out *NTN1*, *NTN3*, *NTN4*, *NTN5*, the UNC5H receptor homologs (*UNC5A*, *UNC5B*, *UNC5C*, and *UNC5D*), as well as *DCC*, *DSCAM*, and *NEO1* alone or in multigene combinations using lentiviral delivery of sgRNAs and Cas9 (**Fig. 5** - **Fig. Sup. 1A-C**). We assessed the effect of their loss on spheroid survival in six HGSOC cell lines (iOvCa147, TOV1946, OVCAR3, OVCAR4, OVCAR8, and COV318). Loss of at least one Netrin ligand resulted in reduced spheroid cell survival in five of six cell lines (**Fig. 5B**). Notably, Netrin-3 loss showed the most pronounced effect on survival. Cell lines such as OVCAR8 and COV318 were relatively resistant to the deletion of Netrin pathway components as individual UNC5, *DCC*, and *DSCAM* genes rarely affected viability (**Fig. 5B**). In these instances, combined deletions of UNC5 family members, or *DCC;DSCAM;NEO1* (DDN for short) were either inviable or highly sensitive to suspension culture induced cell death, suggesting redundancy in this pathway in some cell lines. Other cell lines, such as OVCAR3, were highly sensitive to the loss of multiple individual ligands and any individual UNC5H receptor was highly sensitizing to death induction in suspension (**Fig. 5B**). Overall, these illustrate a robust role for Netrins in survival in this dormant cell culture assay. Importantly, loss of receptor expression does not elevate viability as expected for a dependence receptors, nor does it induce DAPK1 dephosphorylation (**Fig. 5** - **Fig. Sup. 1D**), suggesting that Netrin receptors in HGSOC spheroids transmit positive survival signals.

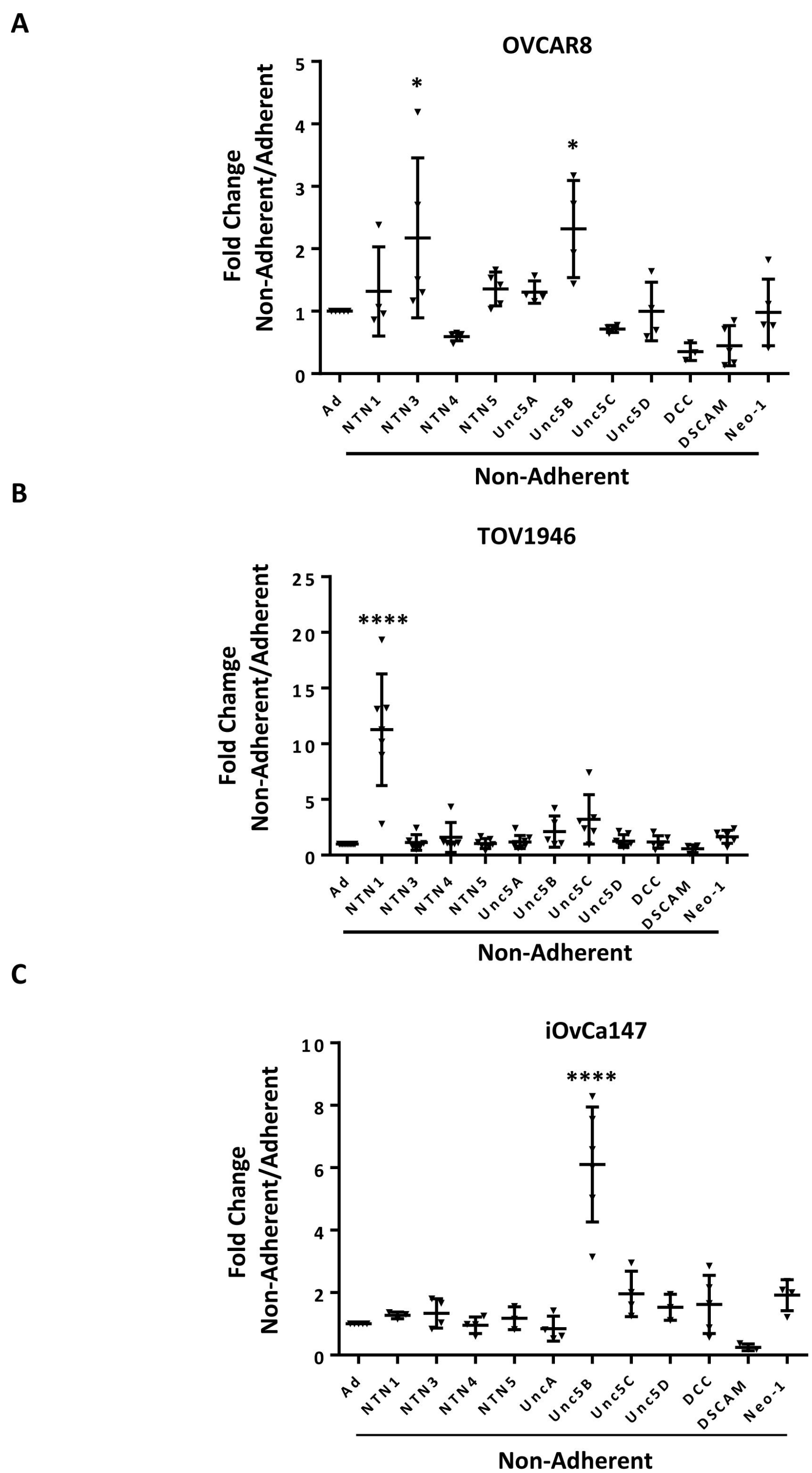

Figure 3.

Expression of Netrin ligands and their dependence receptors is increased in suspension culture.

A-C) RT-qPCR was performed to quantitate mRNA expression levels of Netrin ligands and receptors in three different HGSOC cell lines. Relative expression of the indicated transcripts is shown for suspension culture conditions compared with adherent. All experiments were performed in at least triplicate biological replicates. Means were compared with the same gene in adherent culture using a one way anova (*P<0.05, ****P<0.0001).

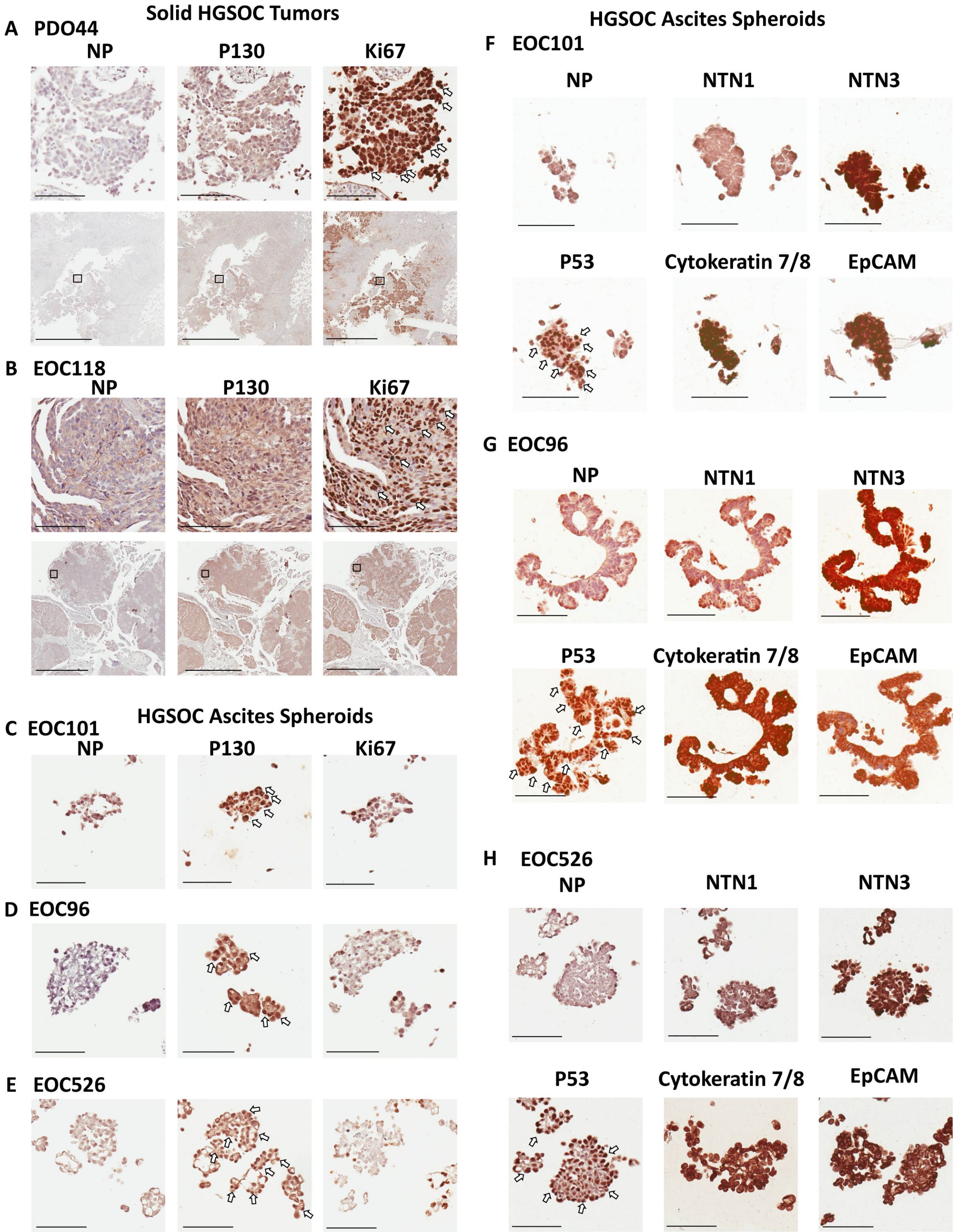

Figure 4.

Netrins are expressed in dormant patient derived spheroids.

Spheroids were isolated from ascites of HGSOC patients and processed for immunohistochemical staining. **A-B)** Serial sections from solid HGSOC tumors were stained with antibodies to p130, Ki67, or a no primary antibody control (NP). Scale bar = 100µm for upper panels and 2mm for lower panels. **C-E)** Serial spheroid sections from the indicated patients were stained with Ki67 and p130. Sections with dense positive nuclear staining are indicated with white arrows. Scale bar = 100µm. **F-H)**. Immunohistochemical staining was performed for the indicated proteins on serial sections of ascites derived spheroids. Omission of primary antibody was used a control for background staining for each patient sample (NP). Scale bar = 100µm

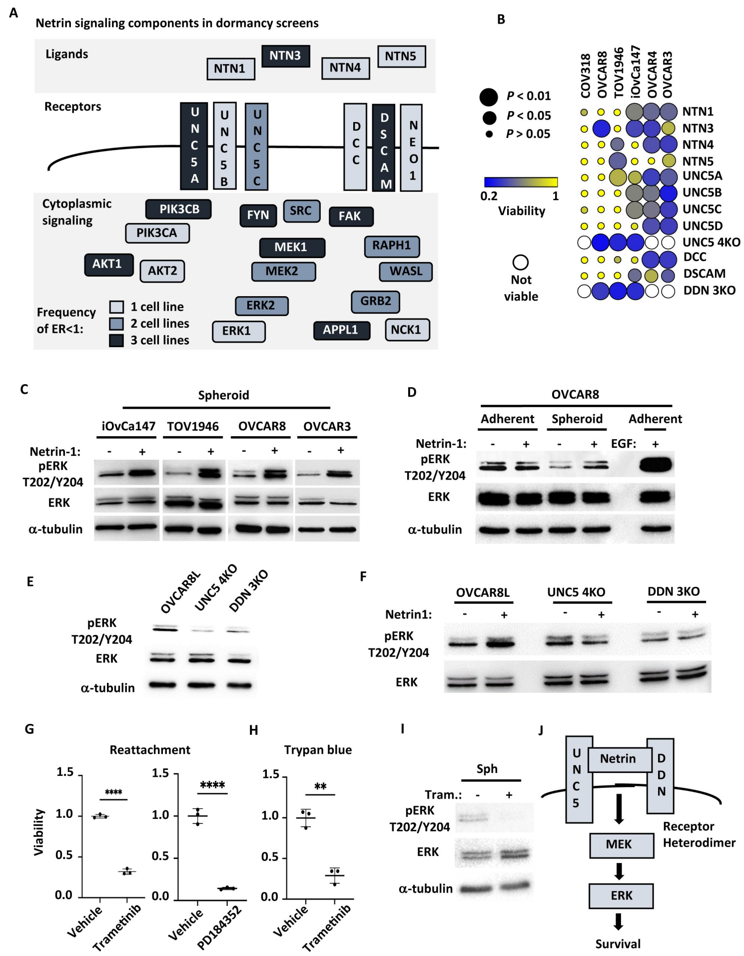

Figure 5.

Netrin ligands and their receptors are required for spheroid cell survival.

A) Illustration of netrin ligands, receptors, and other intracellular signaling molecules that are included in the Axon Guidance pathway category. The frequency of their identification in CRISPR screens is illustrated by shading and indicates how many cell lines had an enrichment ratio <1 for a given component. **B)** The indicated ovarian cancer cell lines were infected with lentiviruses expressing sgRNAs directed against the indicated Netrin signaling genes. Cells were transferred to suspension culture conditions to induce spheroid formation for 72 hours and then returned to adherent conditions for 24 hours to facilitate reattachment. Re-attached cells were stained with Crystal Violet and retained dye was extracted and quantitated to measure relative survival. Each cell-gene combination was assayed in at least three biological replicates, averaged, and viability is displayed as a bubble plot. Mean survival for a given cell-gene combination was compared with GFP control gRNA transduced cells using one way anova and significance levels are illustrated by bubble size. Inviable cell-gene combinations are depicted as empty spaces. **C)** Cultures of the indicated cell lines were cultured in suspension for five days and stimulated with 0.5 μg/mL Netrin-1. Netrin-1 signaling was analyzed by SDS-PAGE and western blotting for phospho-ERK, ERK, and tubulin. **D)** Quiescent adherent OVCAR8 cells were stimulated with 0.5 μg/mL Netrin-1 or 0.5 μg/mL EGF, and compared with OVCAR8 cells in suspension stimulated as in C. Extracts were prepared and blotted for phospho-ERK, ERK, and tubulin. **E)** Suspension cultures of OVCAR8, or knock out derivatives, were harvested and analyzed for relative phosphorylation levels of ERK by western blotting. Total ERK and tubulin blotting serve as expression and loading controls. **F)** Netrin-1 signaling in OVCAR8, UNC5 4KO and DDN 3KO derivatives was tested by transferring cells to suspension and stimulating with Netrin-1 as before. Western blotting for phospho-ERK, ERK were also as before. **G)** OVCAR8 cells were seeded in suspension culture and treated with the MEK inhibitors PD184352, Trametinib or DMSO vehicle for 72 hours. Mean viability was determined by re-attachment and compared by one way anova (****P<0.0001). **H)** OVCAR8 cells were cultured in suspension and treated with Trametinib or DMSO vehicle as in G. Viability was determined by trypan blue dye exclusion and compared by one way anova (**P<0.01). **I)** Extracts were prepared from Trametinib and control treated spheroid cells and blotted for phospho-ERK, ERK, and tubulin. **J)** Model summarizing the roles of Netrin ligands, receptors, and downstream targets MEK and ERK in dormant survival signaling.

To determine if Netrin dependent signaling is active in HGSOC spheroids, we stimulated cells in suspension with recombinant Netrin-1 (**Fig. 5C**). Western blotting for phospho-ERK, as a measure of activation, revealed stimulation of MEK-ERK signaling in each of the cell lines tested. Because MAPK signaling is best known for its role in proliferation, we also tested Netrin-1 stimulation of quiescent, adherent cells and compared this with stimulation in spheroid culture. Both conditions display similar, low levels of phospho-ERK, but Netrin-1 was only able to stimulate ERK phosphorylation in suspension (**Fig. 5D**). Importantly, stimulation of adherent cells with an equivalent concentration of EGF induced pERK levels commonly seen in proliferating cells. Collectively, the context and magnitude of pERK stimulation by Netrin-1 is inconsistent with proliferative signaling, suggesting it may contribute to survival. This suggested that Netrin is a candidate pathway to provide low level, but essential survival signals through the MAPK pathway in dormant spheroids. We investigated the dependence of ERK phosphorylation on UNC5H and DDN receptors and discovered that it is lower when receptors are deleted (**Fig. 5E**). Furthermore, Netrin-1 stimulation of suspension cultures of OVCAR8 cells, or derivatives deleted for UNC5 family members or DDN receptors, indicated that Netrin receptors are essential to activate ERK in suspension culture (**Fig. 5F**). To confirm that Netrin-MEK-ERK signaling contributes to survival in suspension culture, we used two different MEK inhibitors, PD184352 and Trametinib, to treat OVCAR8 suspension cultures and observed that it causes loss of cell viability in suspension culture (**Fig. 5G**). Since we routinely use reattachment of spheroids to adherent plastic as a test of viability, also tested viability using trypan blue dye exclusion to confirm that Trametinib is truly killing cells in suspension and it revealed the same conclusion (**Fig. 5H**). As expected, treatment of cells in suspension with Trametinib lead to lower phospho-ERK levels (**Fig. 5I**). The simplest explanation of this data is that Netrins signal through a heterodimer or multi-receptor complex containing both an UNC5 and DDN component that provides low level stimulation for MEK and ERK specifically in dormant culture conditions.

Netrin signaling is essential for dormant spheroid survival during metastatic dissemination

To investigate the role of Netrin signaling in the context of residual disease and relapse, we xenografted mice with control OVCAR8L cells, or UNC5 4KO cells, by intraperitoneal injection (**Fig. 6A**). Two weeks following injection, mice were euthanized, and spheroids were obtained by washing the abdominal cavity with PBS. Importantly, at this early stage of engraftment, solid tumor lesions were not yet observable, indicating that we are modeling minimal residual disease where dormancy is most relevant. Extraction of RNA from control OVCAR8 cell spheroids recovered at this time point indicates a reduction in expression of pro-proliferative genes compared to the same cells prior to engraftment (**Fig. 6B**). Furthermore, comparison with RT-qPCR measurements of these same markers in suspension culture induced OVCAR8 spheroids indicates that spheroids from abdominal washes are equivalently withdrawn from the cell cycle (**Fig. 6**, **- Fig. Sup. 1**). This suggests spheroids at this stage are dormant. Furthermore, recovery of spheroids by replating on adherent plastic indicates that deletion of UNC5 receptors to block Netrin signaling extensively reduces survival under these circumstances (**Fig. 6C**). We also quantitated cancer cells in these abdominal washes by qPCR using human Alu repeat detection to count cells and this also demonstrates that cancer cell abundance is reduced in this dormant state by loss of UNC5 receptors. Further xenografts were performed to investigate the long term consequences of reduced survival of UNC5 4KO cells in mice. This demonstrated that mice engrafted with UNC5 4KO cells had a considerably longer survival than control OVCAR8 recipient mice that received the same number of cells (**Fig. 6E,F**). Collectively, this demonstrates that inhibition of Netrin signaling compromises viability and eventual disease progression in a model of dormancy and dissemination.

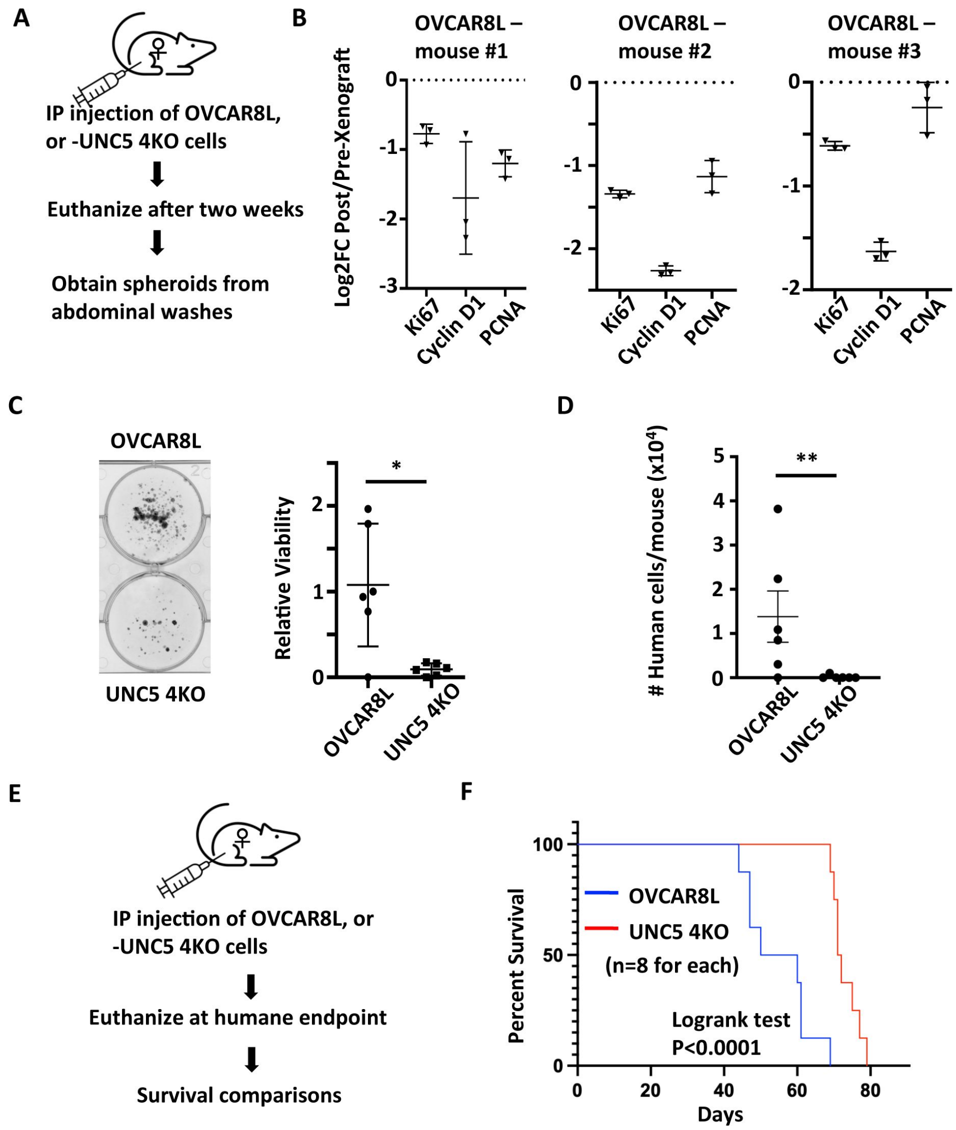

Figure 6.

Loss of Netrin signaling reduces spheroids and prolongs survival.

A) Control OVCAR8L (negative control CRISPR targeted cells) and UNC5 4KO cells were injected into the peritoneal space of female NOD/SCID mice. After two weeks, animals were euthanized and engrafted cells were collected with abdominal washes. **B)** RT-qPCR was used to compare gene expression of the indicated cell cycle markers in RNA extracted from proliferating cells in culture before engraftment and compared with RNA obtained from spheroids in xenografted mice. Technical replicates of RNA derived from three different mice is shown. **C)** Spheroids obtained from abdominal washes from each mouse were plated in adherent conditions to compare their abundance between control and UNC5 4KO genotypes. One example of each genotype of cell is shown. Crystal violet staining and dye extraction were used to quantitate biomass and averages were compared by one way anova (n=6, *P<0.05). **D)** DNA was extracted from cells collected in abdominal washes and human Alu repeats were detected by qPCR to quantitate and compare human cancer cells. Means were compared by one way anova (n=6, **P<0.01). **E)** Control OVCAR8 and UNC5 4KO cells were xenografted as above and mice were monitored until humane endpoint. **F)** Kaplan-Meier analysis of survival for mice engrafted with the indicated genotypes of cells. Survival was compared by logrank test.

Netrin overexpression is correlated with poor outcomes in HGSOC

Genomic studies indicate that HGSOC is one of the most aneuploid cancers (38). In addition to the frequency of copy number variants per tumor, the locations of gains and losses are dissimilar between cases, making this a highly heterogenous disease. Evaluation of TCGA genomic data for HGSOC indicates rare amplifications or deletions of Netrin or Netrin-receptor genes (**Fig. 7 - Fig. Sup. 1A**). Because of redundancy of Netrins and their receptor families this data suggests HGSOC cases with deficiency for Netrin signaling are rare. This is consistent with our data that all six cell lines tested reveal evidence of Netrin component dependence for survival (**Fig. 5B**). Some TCGA HGSOC cases display elevated gene expression for Netrins or their receptors, but only rare cases display low expression (**Fig. 7 - Fig. Sup. 1B**). This suggests that Netrin signaling in HGSOC is retained despite high copy number variation in this cancer type and there is occasional overexpression.

In agreement with our experimental data that multiple Netrin ligands can contribute to spheroid survival in HGSOC (**Fig. 5A&B**), sorting cases by high level expression of Netrin-1 and -3 ligands reveals a significantly shorter overall survival compared to cases with low level expression and this trend is also evident in the cases that only overexpress Netrin-3 (**Fig. 7A**). This genomic data supports the interpretation that Netrin signaling is retained in HGSOC and its elevation is deleterious to patient outcome.

We investigated overexpression of Netrin-1 and -3 in spheroid survival by generating OVCAR8 derivatives that stably overexpress these proteins (**Fig. 7B**). We then utilized these cells in suspension culture assays for spheroid survival as before (**Fig. 7C**). This data indicates that overexpression of either Netrin-1 or -3 increases OVCAR8 cell survival in dormant spheroid culture conditions. Examination of fixed spheroids isolated from this experiment further indicates that elevated survival in this experiment is explained by an increase in abundance of individual spheroids that overexpress either Netrin-1 or -3 (**Fig. 7D**). These experiments reveal the significance of a previously undiscovered role for Netrins in HGSOC. High level expression correlates with poor outcome and in suspension culture, overexpression confers oncogenic properties on HGSOC cells.

Netrin overexpression induces abdominal spread of HGSOC

Our findings from CRISPR screens and cell culture experiments indicate that Netrin signaling supports dormant HGSOC cell survival. Our analysis of TCGA is also predictive of a worse outcome for patients. We sought to solidify these findings by searching for how Netrin signaling affects HGSOC disease pathogenesis.

We utilized a xenograft assay in which OVCAR8 cells overexpressing Netrin-1 or -3 were injected into the peritoneal cavity and compared with OVCAR8 cells bearing an empty vector as control. Mice were harvested following 35 days to analyze disease dissemination and characterize the effect of Netrins in this model of HGSOC metastatic spread (**Fig. 8A**). Necropsy of all animals at endpoint allowed for the identification and quantitation of tumor nodules within the abdomen (**Fig. 8B**). Petal plots illustrate the frequency of specific organ and tissue sites displaying tumor nodules (**Fig. 8B**). This revealed that spread to the omentum was similar across all genotypes of xenografted cells and it occurred in all mice (**Fig. 8B**). In contrast, tumor nodules were detected with increased abundance on the diaphragm, the liver, and mesenteries that support the uterus (mesometrium) (**Fig. 8B**). In addition to more frequent disease spread to these locations caused by Netrin overexpressing cells, we note that the quantity of nodules was also increase in comparison with OVCAR8 control cells (**Fig. 8C-F**). We confirmed the source of these lesions to be the xenografted cells using immunohistochemical staining for human cytokeratin (**Fig. 8G,H**). Lastly, Netrin-1 and -3 histochemical staining confirms its presence in the local tumor

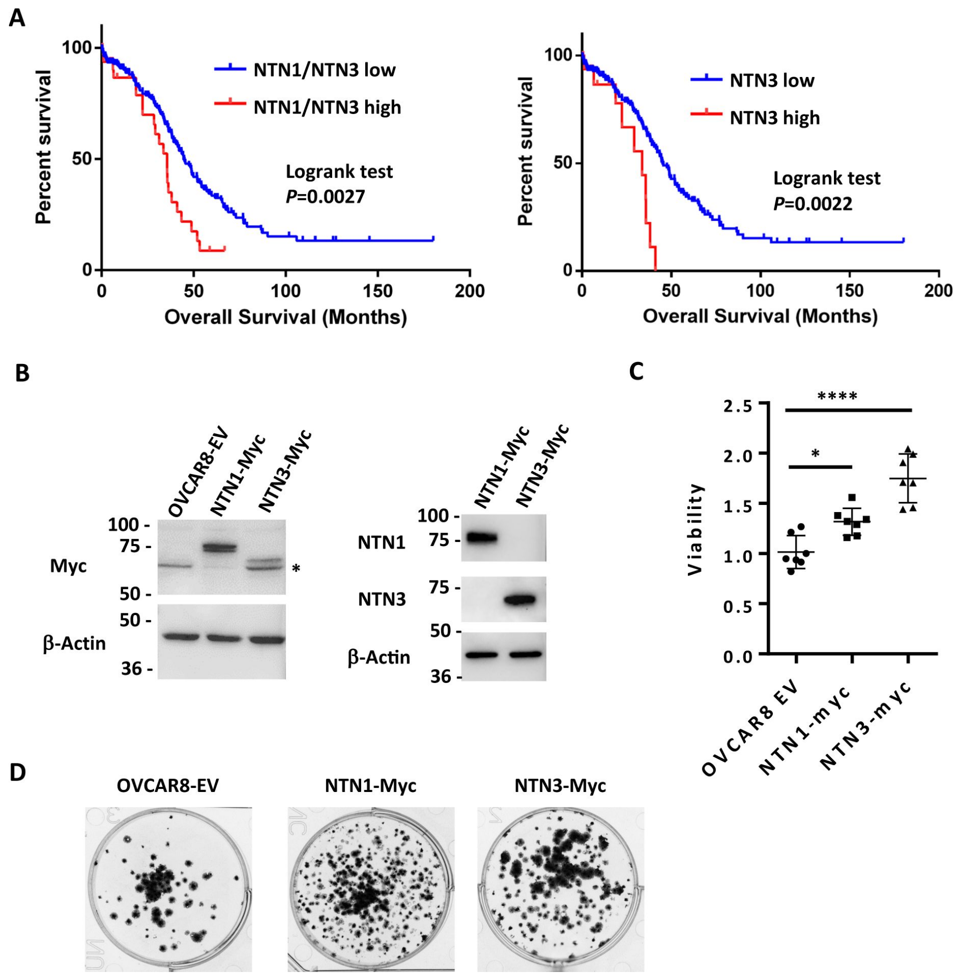

Figure 7.

Netrin ligand overexpression is associated with poor clinical outcome in HGSOC.

A) TCGA RNA-seq data for HGSOC patients (TCGA PanCancer Atlas study) was used to identify high Netrin-1 or -3 and low Netrin-1 or -3 expressing patients (high expressing are above z-score 1.2). Overall survival was used to construct Kaplan-Meier plots and survival was compared using a logrank test. **B)** OVCAR8 cells were stably transduced with lentiviral constructs to overexpress epitope tagged Netrin-1 or -3. Western blotting for Netrins, Myc-tags, and Actin were used to determine relative expression levels of both Netrins in these cell populations and vector controls. **C)** Control and Netrin overexpressing cells were transferred to suspension culture conditions to form spheroids, and replated to assay for viability. Mean viability and standard deviation is shown for each. One way anova was used to compare survival (* P<0.05, *** P<0.001). **D)** Reattached spheroids were fixed and stained with Crystal Violet to examine size and abundance in control and Netrin-1 or -3 overexpression.

microenvironment (**Fig. 8G,H**). The similarity of increased tumor nodules seen in xenografts and elevated spheroid formation and viability seen in culture suggests that Netrin signaling improves cell viability in spheroid aggregates in vivo leading to increased disease spread.

Discussion

In this report we investigated survival mechanisms in a model of HGSOC dormancy. We utilized a modified CRISPR screening approach that was adapted to discover gene loss events in largely non-proliferative conditions. We compared essential genes discovered in the screen with gene expression increases identified by RNA-seq in dormant HGSOC cultures. This implicated the Netrin signaling pathway in HGSOC spheroid survival. We independently validated the role of Netrins, their receptors, and downstream intracellular targets MEK and ERK in the survival biology of HGSOC cells in dormant culture and in vivo in xenografts. Furthermore, we reveal that overexpression of Netrin-1 and -3 are correlated with poor clinical outcome in HGSOC and over expression of these ligands induces elevated survival in dormant HGSOC culture conditions and increased disease spread in a xenograft model of dissemination. Our work implicates Netrins, their receptors, and MEK-ERK as a previously unappreciated, but critical signaling network in HGSOC survival.

HGSOC spheroid cell biology has presented a unique challenge to cancer chemotherapy. Dormant spheroids are most abundant in patients with late-stage disease and they contribute to therapeutic resistance and seed metastases (39–41). Hence, there is a critical unmet need to understand HGSOC spheroid dependencies and target them therapeutically. Our data suggests that Netrin signaling is a universally active mechanism in spheroid cell survival. HGSOC is characterized by extensive genomic rearrangements and copy number variants (38), and we speculate that Netrin signaling is retained because of the extensive redundancy of components of this pathway. Netrin-1 and -3 can both contribute to survival signaling and knock out of three or four different Netrin receptors is necessary to compromise Netrin signaling in survival. A key goal for the future is to identify Netrin-1 or -3 overexpression patients and compare their specific disease progression patterns with our xenograft model. This will be most informative of the effects of Netrin ligand overexpression in scenarios of dormancy, metastasis, and chemotherapy resistance.

This study demonstrates that Netrins function to stimulate UNC5H and fibronectin repeat containing receptors such as DCC to activate survival signals in HGSOC spheroids. This is distinct from the known role of UNC5H and DCC as dependence receptors where Netrin inhibits their pro-apoptotic signaling (25). Our data does not show elevated survival of spheroid cells in response to deletions of receptors, even in instances where families of Netrin receptors are co-deleted. Consistent with this observation, over expression of Netrin-1 or -3 in OVCAR8 cells stimulates higher level survival in suspension. In colon cancer, deletion of DCC confers a survival advantage and underscores the dependence receptor concept (42, 43). In HGSOC DCC or UNC5H deletions are relatively rare (38). This suggests that ovarian cancer cells utilize Netrin signals for positive survival signaling instead of to neutralize inherent death signals from dependence receptors. Our data suggests that Netrin activates MEK and ERK to support cell survival indicating that targeting ligands, receptors or downstream at MEK are all possible approaches to inhibit this survival pathway. Netrin-1 inhibition in solid tumor xenograft models of melanoma have been shown to be effective in combination with chemotherapy (44), but its effects on a dormant scenario are unknown. Identifying agents that cooperate with standard HGSOC chemotherapeutic agents such as carboplatin would offer an attractive strategy to eradicate dormant cells while combating progressive disease. Alternatively, a novel approach to targeting dormancy discovered in this study is, ironically, MEK inhibition. While MAPK signaling in cancer is universally known for its role in proliferation (45), and MEK inhibitors are used to block proliferation (46, 47), it is noteworthy that Netrin stimulation of MAPK signaling is reported to provide survival signals and to guide axon outgrowth in a non-proliferative context (26, 48). Dormancy is

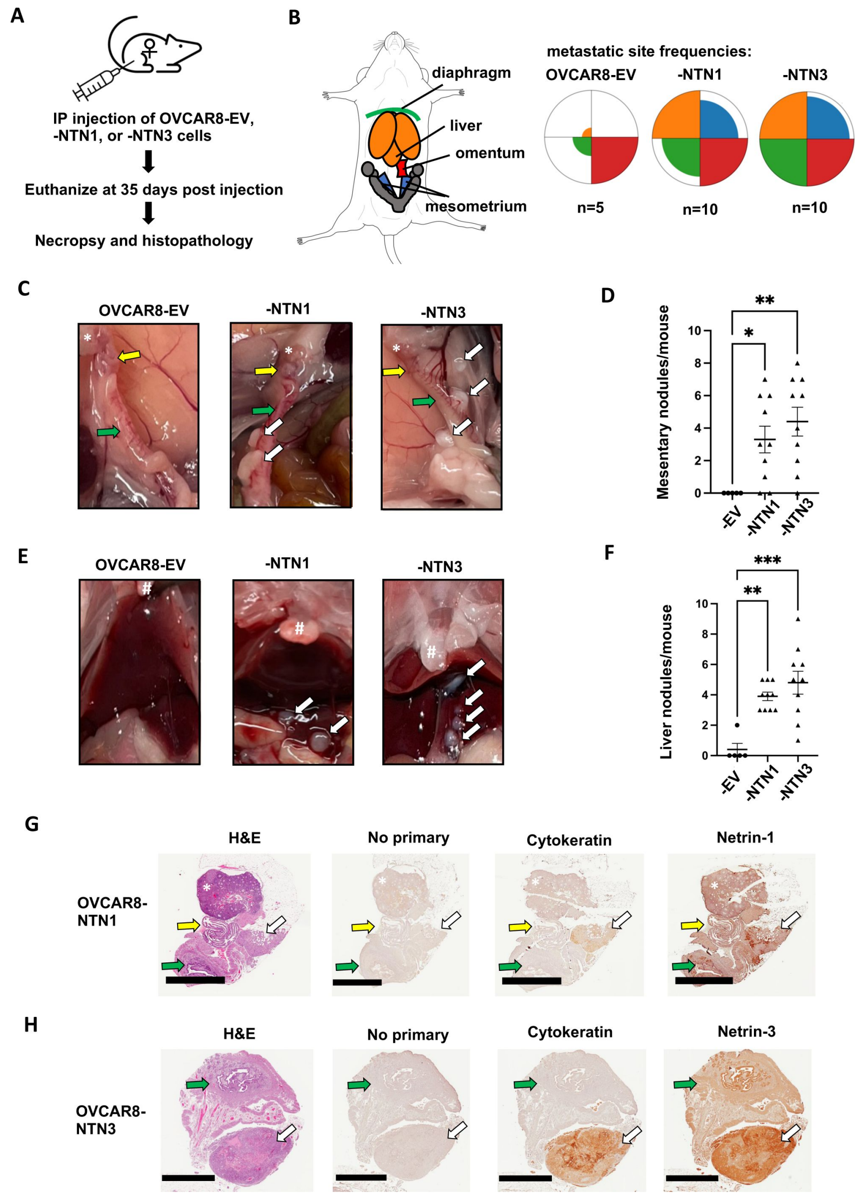

Figure 8.

Netrin overexpression causes increased dissemination of tumor nodules.

A) Netrin overexpressing and control OVCAR8 cells were injected into the intraperitoneal space of female NOD/SCID mice. Mice were euthanized following 35 days and analyzed for disease burden by necropsy and histopathology. **B)** The spread of cancer to the diaphragm, liver, omentum, and mesometrium was determined from necropsies and colors used in the anatomical schematic correspond with petal plots for each genotype of cells. Petal plot radius illustrates frequency of mice bearing disease spread to a particular location. The radius of color fill is proportional to the total number of animals with tumor nodules found in that location. **C)** Photographs of necropsy findings in the mesometrium. Locations of ovaries (*), oviduct (yellow arrow), and uterine horn (green arrow) are indicated in each case. Tumor nodules are indicated by white arrows. **D)** The number of mesometrium associated tumor nodules was determined for each mouse. Mean values are indicated and differences between genotype were determined by one way anova (* P<0.05, ** P<0.01). **E)** Photographs of necropsy findings in the liver. Location of sternum is indicated (#) in each image. Tumor nodules are indicated by white arrows. **F)** The number of liver associated tumor nodules was determined for each mouse. Mean values are indicated and differences between genotype were determined by one way anova (** P<0.01, *** P<0.001). **G** and **H)** Histology of mesometrial tumor nodules are shown. Serial sections were stained with H&E, or with the indicated antibodies for immunohistochemistry. Ovaries are indicated (*), as are the oviduct (yellow arrow), the uterus (green arrow), and the tumor nodule (white arrow). Scale bar = 2mm characterized by low phospho-ERK levels (3, 6), indicating low MEK activity, however low levels of RAS-MAPK signaling are known to provide survival signals (45). A challenge with deploying MEK inhibitors as an anti-dormancy setting in HGSOC therapy is its incompatibility with standard chemotherapies that rely on cell proliferation. Overall, our study aimed to identify potential new therapeutic targets in dormancy and Netrins and MEK are potential new candidates for dormant HGSOC disease.

Current literature offers little to connect Netrins with HGSOC beyond the observation that *NTN1* expression is increased in tumors compared to benign lesions (49). Similarly, dormancy literature does not include a role for Netrins (3–6, 50). However, we expect that Netrin signaling may play an unappreciated role in dormancy in other cancer disease site paradigms. As stated in the introduction, dormancy is characterized by acquisition of stem like properties, EMT, and in most paradigms it involves rare cells establishing themselves in perivascular or bone marrow niches and Netrins have also been implicated in these processes (51–55). Inhibition of Netrin-1 with NP137 and treatment with decitabine has been shown to reduce lung seeding suggesting that Netrins are relevant in a metastatic scenario that may include dormant intermediate steps in dissemination (56). Furthermore, Netrin signaling through UNC5B is known to direct vascular branching (57). This suggests that Netrins are available in the perivascular niche where dormant cells often reside. Lastly, Netrin-1 has been shown to be a component of the bone marrow niche that suppresses hematopoietic stem cell proliferation (51). While not a cancer dormancy scenario, it indicates that Netrins support some solitary dormant cell properties and are expressed by resident cells in the appropriate dormant cell niches beyond just HGSOC. For these reasons we expect Netrins will have a significant role in tumor dormancy beyond ovarian cancer.

Methods

Cell Lines and Engineering Cas9+ cells

High-grade serous ovarian cancer (HGSOC) cell lines OVCAR3, OVCAR4, OVCAR8, COV318, TOV1946 and iOvCa147 were used in this study. The iOvCa147 cells have previously been characterized and reported (58). All cell lines were verified by STR analysis. iOvCa147, OVCAR8, and TOV1946 cells were engineered to express Cas9 using pLentiCas9-Blast and clones were isolated. High Cas9 editing efficiency was determined by viability studies using sgRNAs targeting selected fitness genes (PSMD1, PSMD2, EIF3D) and a non-targeting control (LacZ) as previously described (59).

Lentiviral sgRNA library preparation

HEK293T cells were transfected with the combined A and B components of the GeCKO v2 whole genome library (123,411 sgRNAs in total) along with plasmids encoding lentiviral packaging proteins (60). Media was collected 2-3 days later, cells and debris were pelleted by centrifugation, and supernatant containing viral particles was filtered through a 0.45 μM filter and stored at −80°C with 1.1 g/100 mL BSA.

GO-CRISPR screens in iOvCa147, OVCAR8, and TOV1946 cells

Cas9-positive and Cas9-negative cells were separately transduced with lentiviruses at a multiplicity of infection of 0.3 and with a predicted library coverage of >1000-fold. Cells were selected in puromycin and maintained in complete media (DMEM/F12 with 10% FBS, 1% pen-strep-glutamine), containing puromycin in all following steps. More than 10^9 cells were collected and split into three groups consisting of approximately 3.0×10^8 cells. Triplicate samples of 6.2×10^7 cells were saved for sgRNA sequence quantitation (T_0). The remaining cells were seeded at 2.0×10^6 cells/mL in 20 x 10 cm ULA plates. Following 2 days of culture, media containing spheroids was transferred to 10 x 15 cm adherent tissue culture plates. The next day unattached spheroid cells were collected and re-plated onto additional 15 cm plates. This process was repeated for a total of 5 days to maximally capture viable cells. Attached cells were collected and pooled for DNA extraction (T_f). Genomic DNA was extracted using the QIAmp DNA Blood Maxi Kit (Qiagen, #51194). PCR amplification and barcoding performed as described (Hart paper, Cell, 2015) using NEBNext Q5 Ultra mastermix (NEB, # MO544). PCR products were extracted from agarose using a Monarch DNA Gel Extraction Kit (NEB, # T1020), quantitated and next generation sequencing was performed using an Illumina NextSeq 75 cycle High Output kit on an Illumina NextSeq platform.

TRACS analysis

The library reference file containing a list of all sgRNAs and their sequences (CSV file), raw reads for the pooled sgRNA library (FASTQ files (L0) and raw reads (FASTQ files) for all T_0 and T_f timepoints and replicates for Cas9-positive and Cas9-negative cells were loaded into TRACS (*https://github.com/developerpiru/TRACS*)(61). TRACS then automatically trimmed the reads using Cutadapt (v1.15), built a Bowtie2 (v2.3.4.1) index and aligned the reads using Samtools (v1.7). TRACS used the MAGeCK read count function (v0.5.6) to generate read counts and incremented all reads by 1 to prevent zero counts and division-by-zero errors. The TRACS algorithm was then run using this read count file to determine the Library Enrichment Score (ES), Initial ES, Final ES and the Enrichment Ratio (ER) for each gene. VisualizeTRACS was used to generate graphs of Initial and Final ratios.

Pathway enrichment analyses

For GO-CRISPR screens and multi-omics comparisons, we used the filtered list of genes obtained using TRACS and performed gene ontology and pathway enrichment analysis using the ConsensusPathDB enrichment analysis test (Release 34 (15.01.2019), *http://cpdb.molgen.mpg.de/*) for top-ranked genes of interest. p_{adj} values and ER values for each gene were used as inputs. The minimum required genes for enrichment was set to 45 and the FDR-corrected p_{adj} value cutoff was set to < 0.01. The Reactome pathway dataset was used as the reference. For each identified pathway, ConsensusPathDB provides the number of enriched genes and a q value (p_{adj}) for the enrichment. Bar plots were generated in R 3.6.2 using these values to depict the significant pathways identified.

Generating DYRK1A knockout cells

A double cutting CRISPR/Cas9 approach with a pair of sgRNAs (sgRNA A and B) was used to completely excise exon 2 (322 bp region) of *DYRK1A* using a px458 vector (Addgene #48138) that was modified to express the full CMV promoter. PCR primers that flank the targeted region of *DYRK1A* were used to verify deletion. Single-cell clones were generated by liming dilutions and evaluated for DYRK1A status by PCR, western blot, and sequencing. See Supplemental Table 1 for sgRNA and PCR primer sequences.

DYRK1A IP kinase assay

Whole cell lysates from adherent parental iOvCa147 cells and *DYRK1A*$^{-/-}$cells were extracted using complete RIPA buffer and incubated overnight with DYRK1A antibody (Cell Signaling Technology anti-DYRK1A rabbit antibody #8765). Samples were then washed with buffer and Dynabeads (Thermo Fisher Scientific Dynabeads Protein G #10003D) were added for 2 hours. Samples were then washed with buffer and recombinant Tau protein (Sigma recombinant Tau protein #T0576) and ATP (Sigma #A1852) were added and samples were incubated for 30 minutes at 37°C. 5x SDS was then added and samples were resolved by SDS-PAGE. Antibodies used for western blotting were phosphospecific Tau antibody (Cell Signaling Technology phospho-Tau Ser-404 rabbit antibody #20194) and Tau protein (Cell Signaling Technology anti-Tau rabbit antibody #46687).

RNA preparation and RNA-sequencing

Total RNA was collected from parental iOvCa147 cells and *DYRK1A*$^{-/-}$cells from 24-hour adherent or 6-hour spheroid conditions using the Monarch Total RNA Miniprep Kit (NEB #T2010S) as described above. RNA was collected in three replicates for each condition (adherent and spheroids). Samples were quantitated using Qubit 2.0 Fluorometer (Thermo Fisher Scientific) and quality control analysis using Agilent 2100 Bioanalyzer (Agilent Technologies #G2939BA) and RNA 6000 Nano Kit (Agilent Technologies #5067-1511). Ribosomal RNA removal and library preparation was performed using ScriptSeq Complete Gold Kit (Illumina #BEP1206). High-throughput sequencing was performed on an Illumina NextSeq 500 platform (mid-output, 150-cycle kit).

RNA-seq analyses

Raw FASTQ data was downloaded from Illumina BaseSpace. Reads were aligned using STAR 2.6.1a(62) to the human genome (Homo_sapiens.GRCh38.dna.primary_assembly.fa; sjdbGTFfile: Homo_sapiens.GRCh38.92.gtf) to generate read counts. DESeq2 was run with BEAVR(63) and used to analyze read counts (settings: False discovery rate (FDR): 10%; drop genes with less than 1 reads) to identify differentially expressed genes ($\log_2$ fold change > 1 for upregulated genes; or $\log_2$ fold change < -1 for downregulated genes; $p_{adj} < 0.05$) for the following comparisons: parental iOvCa147 adherent cells vs. parental iOvCa147 spheroid cells; or parental iOvCa147 spheroid cells vs. *DYRK1A*$^{-/-}$ spheroid cells. For RNA-seq, we formed pathway analyses in BEAVR and downloaded the pathway enrichment table to construct bar plots and determine overlapping pathways using R 3.6.2.

Western blots

Adherent cells were washed in 1x PBS and lysed on the plate by the addition of complete RIPA buffer with protease (Cell Signaling Technology #5871S) and phosphatase (Cell Signaling Technologies #5870S) inhibitors and incubated for 5-10 minutes on ice. Spheres were collected by gentle centrifugation at 300 rpm, washed gently in ice cold PBS, centrifuged again and then resuspended in RIPA buffer as above. Samples were then centrifuged at 14,000 RPM at 4°C for 10 minutes. The supernatant liquid for each sample was collected and the pellets re-extracted with RIPA containing the above inhibitor cocktails. Following centrifugation, the supernatants for each treatment were collected and pooled. Protein was determined by Bradford assay. To assess the effect of Netrin 1 on downstream signaling, spheres incubated under non adherent conditions for 5 days were challenged with the addition of 500 ng/ml human recombinant Netrin-1 (R&D Systems #6419-N1-025) for a period of 20 minutes before collection. Lysates were mixed with 5x SDS loading dye buffer and resolved using standard SDS-PAGE protocols. Antibodies used for blotting were p42/44 MAPK (Erk1/2) (Cell Signaling Technology rabbit antibody #4695), Phospho-p44/42 MAPK(Erk1/2) T202/Y204 (Cell Signaling Technology; rabbit antibody #4370), p38 MAPK (Cell Signaling Technology; rabbit antibody #9212), p38 MAPK T180/Y182 (Cell Signaling Technology; rabbit antibody #9211), p130(RBL2, Santa Cruz Biotechnology, rabbit antibody sc-317 discontinued), Ki67 (Abcam rabbit antibody, # ab16667), DYRK1A (Cell Signaling Technology anti-DYRK1A rabbit antibody #8765), Netrin-1 (Abcam rabbit antibody #ab126729), Netrin-3 (Abcam rabbit antibody #ab185200), α-tubulin (Cell Signaling Technology anti-αTubulin rabbit antibody #2125) and β-actin (Millipore Sigma rabbit antibody #A2066).

RT-qPCR analysis of gene expression

Cells from a minimum of three biological replicates were triturated off the plate using Versene (0.5mM EDTA in PBS), collected, counted and 10^6 cells plated onto Ultra-Low Attachment cell culture dishes (Corning) to induce sphere formation. The spheres were collected after 24 h and RNA extracted using the Monarch Total RNA Miniprep Kit (New England Biolabs #T2010S). RNA was extracted from adherent cells directly from the plate. Extracted RNA was quantitated using a NanoDrop and stored at -80° until used. First strand cDNA was generated from isolated RNA using the iScript cDNA Synthesis Kit (Bio-Rad, #1708890) and RT-qPCR performed on the resulting cDNA (diluted 5-fold) using the iQ SYBR Green Supermix (Bio-Rad, #1708882) and following the manufactures instructions. Reactions were performed in 0.2ml non-skirted low profile 96 well PCR plates (Thermo Scientific, #AD-0700) using a CFX Connect Real Time System thermo-cycler (Bio-Rad).

TCGA patient data analyses

We used the cBioPortal for Cancer Genomics (*https://www.cbioportal.org*)(64) to analyze Netrin pathway alterations in Ovarian Serous Cystadenocarcinoma patients (TCGA, PanCancer Atlas) for genomic alterations and mRNA expression. OncoPrints were generated for mRNA expression (mean z-score threshold ± 1.5), and mutations and putative copy number alterations for the following genes: *NTN1, NTN3, NTN4, NTN5, UNC5A, UNC5B, UNC5C, UNC5D, DCC, DSCAM*. Survival analysis of *NTN1*, and *-3* utilized patients from TCGA, PanCancer Atlas dataset and RNA-seq expression (high expression z-score threshold > 1.2 and low is below 1.2).

Generation of gene knockout cell lines

Gibson Assembly (NEB #E2611) was used to clone sgRNAs sequences derived from the GeCKO v2 library per gene of interest (*NTN1, NTN3, NTN4, NTN5, DSCAM, DCC, NEO1, UNC5A, UNC5B, UNC5C, UNC5D,* and *EGFP* control (See Supplemental Table 1 for sgRNA sequences for each)) into lentiCRISPR v2 (Addgene #52961). For individual gene knockouts pooled libraries of up to four sgRNAs were generated. For multiple gene knockouts pooled libraries of two sgRNAs/gene were constructed. To each 20-nucleotide guide (Supplemental table I) was appended the 5' sequence, 5'-TATCTTGTGGAAAGGACGAAACACC-3' and the 3' sequence, 5'-GTTTTAGAGCTAGAAATAGCAAGTTAAAAT-3' (65). A 100 bp fragment was then generated using the forward primer, 5'-GGCTTTATATATCTTGTGGAAAGGACGAAACACCG-3', and reverse primer, 5'-CTAGCCTTATTTTAACTTGCTATTTCTAGCTCTAAAAC-3'. The resulting gel purified fragment was quantitated by Qubit and cloned into the lentiviral vector LentiCRISPR-V2 previously digested with BsmBI (New England Biolabs, # R0739) as described (65). To generate virus particles, HEK293T cells were transfected with the assembled plasmid along with plasmids encoding lentiviral packaging proteins. Media was collected after 2-3 days and any cells or debris were pelleted by centrifugation at 500 x g. Supernatant containing viral particles was filtered through a low protein binding 0.45 µM filter. For each knockout, it was important to ensure that each target cell was transduced with a full complement of guides. To ensure this, the cells were transduced at relatively high MOI. Briefly, 25,000 cells were plated and transduced with fresh HEK293T conditioned media containing virus. While we did not titer the viral preparations for these experiments, previous experience indicated that we could generate sufficiently high virus titers using our protocol. Under these conditions, we found a lack of puromycin sensitive cells following 2-3 days in media containing 2-4 µg/ml puromycin indicating that every cell received at least one virus particle. Multi-gene knock outs were created through multiple rounds of infection with the relevant culture supernatants followed by one round of puromycin selection.

Generation of gene overexpression cell lines

Plasmids encoding Netrin1 and Netrin3 were obtained from Addgene (Ntn1-Fc-His, #72104 and Ntn3-Fc-His, #72105). Netrin1 was PCR amplified using the forward primer 5'-GCTATCGATATCCCAAACGCCACCATGATGCGCGCTGTGTGGGAGGCGCTG-3' and reverse 5'-GCTATCTCTAGAGGCCTTCTTGCACTTGCCCTTCTTCTCCCG-3' and cloned into the EcoRV/XbaI sites of pcDNA3.1/myc-HisA. For lentiviral expression, pcDNA NTN1 was amplified using the primers, forward 5'-CACTGTAGATCTCCAAACGCCACCATGATGCGCGCTGTGTGGGAGGCGCTG-3' and, reverse 5'-ACGCGTGAATTCTTATCAACCGGTATGCATATTCAGATCCTCTTCTGAGAT-3' while NTN3 was amplified from pcDNA NTN3 using, forward 5'-CTCGAGAGATCTGCGGCCGCCACCATGCCTGGCTGGCCCTGG-3' and the same reverse primer as for

NTN1. In this way both constructs carry a C-terminal Myc tag. Both cDNA's were inserted into the BamHI/EcoRI sites of the lentiviral expression vector FUtdTW (Addgene #22478). Construction of lentiviral particle and transduction into the target cells was as described above.

Spheroid formation and reattachment viability assays

For each targeted gene or cDNA expressed in iOvCa147, OVCAR3, OVCAR4, OVCAR8, COV318, or TOV1946 cells, spheroid viability was assayed as follows: Three to ten biological replicate cultures of knockout cells were cultured for 72 hours in suspension conditions using ULA plasticware (2×10^6 cells per well) to allow spheroid formation. Addition of therapeutics (Tramatinib, Selleckchem, #S4484) was made at the time of plating. Spheroids were collected and transferred directly to standard plasticware to facilitate reattachment for 24 hours, or dissociated and mixed with trypan blue. Reattached cells were fixed in 25% methanol in PBS for 3 minutes. Fixed cells were incubated for 30 minutes with shaking in 0.5% crystal violet, 25% methanol in PBS. Plates were carefully immersed in water to remove residual crystal violet and destained in 10% acetic acid in 1x PBS for 1 hour with shaking to extract crystal violet from cells. Absorbance of crystal violet was measured at 590 nm using a microplate reader (Perkin Elmer Wallac 1420).

Xenografts

All animal experiments were approved by the Animal Care Committee at Western University (protocol #2020-039). ARRIVE principles were followed in carrying out xenograft experiments. Female NOD-SCID mice were purchased from JAX (#001303). They were randomly selected prior to being engrafted between six and eight weeks of age with 4 x10^6 OVCAR8 control or modified derivatives. Five to ten mice were housed for 14 days, 35 days, or until humane endpoint for each OVCAR8 or modified derivative. Following euthanasia all mice were subjected to a necropsy to analyze disease spread. Tumor lesions were photographed and quantitated by individuals unaware of the genotypes. Abdominal organs and associated tumor lesions were fixed in 10% formalin overnight and transferred to 70% ethanol and later embedded, sectioned and stained as described below. In cases where abdominal washes were performed, an incision was made in the peritoneal membrane and PBS was injected, the abdominal region was agitated and PBS was withdrawn using a P1000 pipet tip and aliquoted for downstream analysis in cell culture and qPCR.

Patient derived spheroids

Patient derived spheroids were isolated from ascites fluid obtained from debulking surgical procedures. These HGSOC samples were obtained with permission and institutional research ethics oversight (Project ID #115904). Spheroids were prepared by filtering the ascites fluid through a 100μm mesh filter and washing the captured spheroids from the filter with PBS. The spheres were collected by gravity on ice in PBS, washed twice with ice cold PBS and allowed to settle on ice. Spheres were then fixed for 20 minutes in 10% formalin, washed, and resuspended in PBS before being embedded in paraffin.

Histology and immunohistochemistry

Fixed organs from xenografts or spheroids were embedded in paraffin, sectioned, and stained with H&E using standard methods. For immunohistochemistry, sections were deparaffinized by standard procedures and antigen retrieval was performed in 10mM citrate, pH6.0. The sections were blocked with 10% goat serum/3% BSA/0.3% Triton X-100 and stained with antibodies to Netrin-1 (Abcam #ab126729, 1:200), Cytokeratin (human specific cytokeratin 7 and 8, ZETA Corporation #Z2018ML, 1:200), EpCAM (Cell Signaling Technology, #14452, 1:250), p53 (Cell Signaling Technologies, #2527, 1:125), Netrin-3 (1:250), Ki67 (Cell Signaling Technology, #12202, 1:400) and p130 (RBL2, Santa Cruz Biotechnology, SC-317, 1:150). Netrin-3 antibodies were generated using immunized rabbits and their generation and purification will be published elsewhere. Primary antibodies were followed with a biotinylated anti-rabbit (Vector Labs, # BA-1000) or anti-mouse (Jackson ImmunoResearch, #115-067-003) IgG as a secondary antibody (1:500). Proteins were visualized using DAB chemistry (ImmPACT DAB, Vector Labs, #SK-4105).

Statistical analyses

Specific statistical tests used are indicated in the figure legends for each experiment. Analysis was done using GraphPad Prism 6.

Materials availability

All previously unpublished reagents created in this study are available on request from the corresponding author.

Data availability

All sequencing data have been deposited in GEO under accession codes GSE190294 and GSE150246. Source data for graphs and original blots are available in source data files, or from public databases as described in the methods section.

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A

TRACS (Toolset for the Ranked Analysis of GO-CRISPR Screens) algorithm:

All Enrichment Scores (ES) for all genes → **Discard genes with Library ES in bottom quartile** → **Compare Spheroid ES and Adherent ES** → **Select genes with negative log2 Spheroid/Adherent ratio, *P*<0.05**

B

Gene scoring formula

Gene score: $$GS_j = \sum_{i=1}^{s} \left(\log_2 \frac{[normalized\ sgRNA\ abundance]_{Cas9+}}{[avg\ normalized\ sgRNA\ abundance]_{Cas9-}} \right)$$

Enrichment score: $$ES_j = \frac{\frac{1}{n}\sum_{i=1}^{n} GS\ rank}{s}$$

Enrichment ratio: $$ER_j = \log_2 \frac{ES_{j_{final}}}{ES_{j_{initial}}}$$

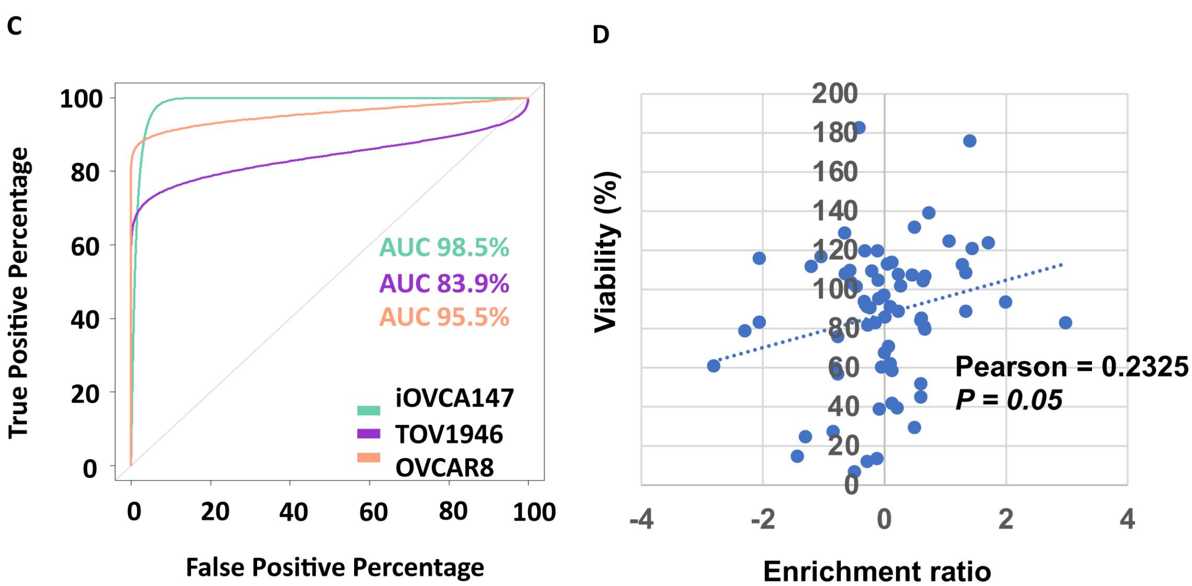

Figure 1 - Figure Supplement 1.

CRISPR screen analysis details and internal controls.

A) Flowchart of data analysis used to identify genes that are most relevant to ovarian cancer cell survival in suspension. **B)** Mathematical formulas to define gene scores, enrichment scores (n=guides per gene), and enrichment ratios used by TRACS. **C)** Reader-operator curves were generated to assess TRACS categorization of 1000 non-targeting control sgRNAs in the GeCKOv2 library for each of the three cell lines screened. **D)** Spheroid cell viability levels from low throughput siRNA gene knock downs for 35 genes in iOvCa147 and OVCAR8 cells was plotted against the enrichment ratio for the same genes in the same two cell lines from this screen. These independent approaches to measure viability were compared using linear regression.

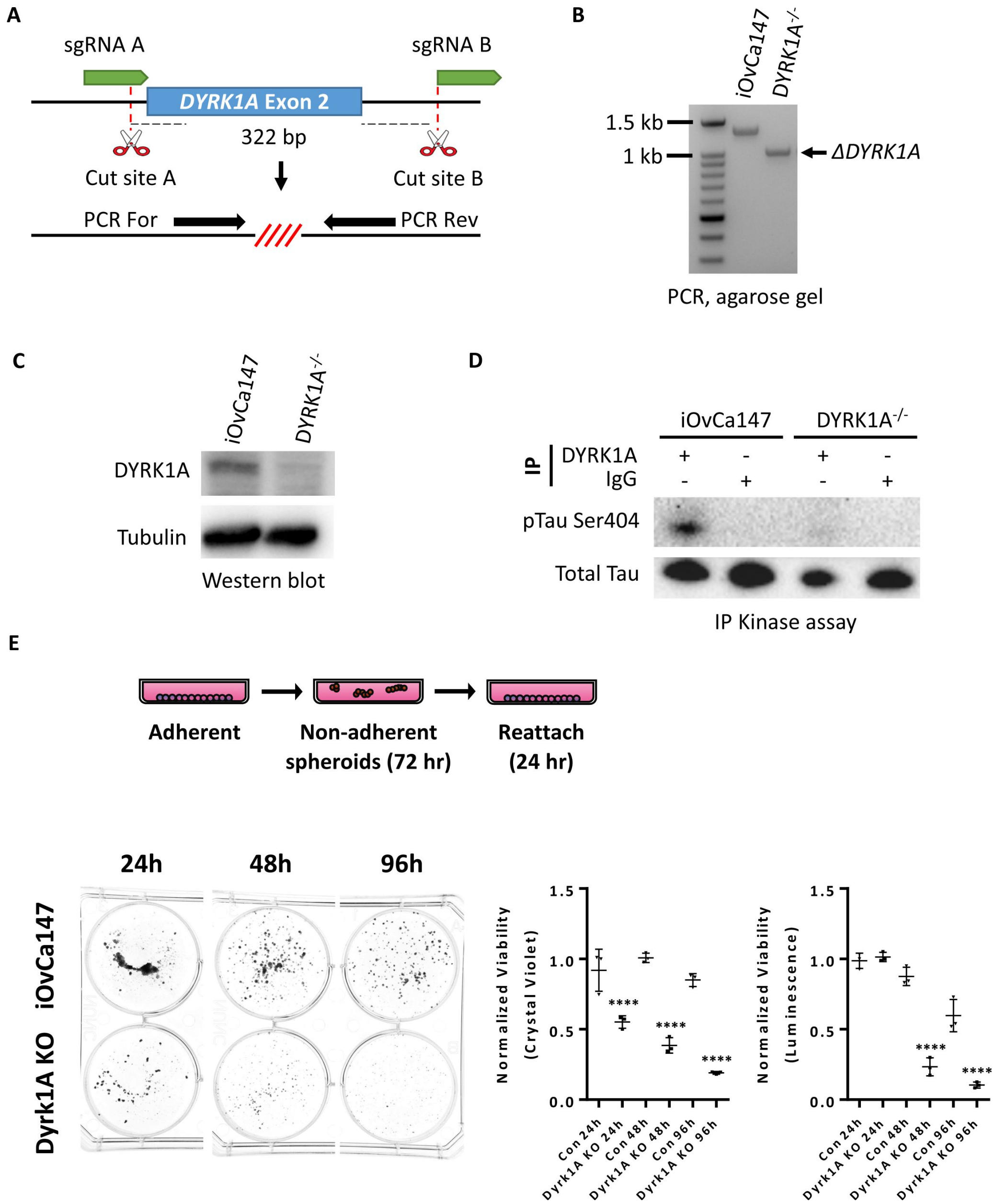

Figure 2 - Figure Supplement 1.

Generation of iOvCa147 cells deficient for DYRK1A and loss of viability in suspension.

A) Strategy to generate *DYRK1A$^{-/-}$* cells using a pair of sgRNAs that flank exon 2. PCR For and PCR Rev primers flank exon 2 and are used to detect deletion events. **B)** Agarose gel showing PCR products for iOVCA147 cells and putative *DYRK1A$^{-/-}$* cells. The full length amplicon containing exon 2 was detected in parental iOvCa147 cells (1,348 bp). A smaller amplicon (1,026 bp) was detected in *DYRK1A$^{-/-}$* cells, indicting successful excision of the 322 bp region encompassing exon 2. **C)** Western blot comparing DYRK1A expression in iOVCA147 cells and *DYRK1A$^{-/-}$* cells. **D)** IP kinase assay to evaluate DYRK1A activity in *DYRK1A$^{-/-}$* cells. Anti-DYRK1A antibodies or IgG was used to immunoprecipitate from iOVCA147 and *DYRK1A$^{-/-}$* cells. Precipitates were incubated with ATP and Tau protein. Samples were resolved by SDS-PAGE and probed with pSer404-Tau antibody. **E)** Parental iOvCa147 or *DYRK1A$^{-/-}$* cells were incubated in suspension conditions for 24 hours, 72 hours, or 4 days to induce spheroid formation, and then re-plated in adherent conditions for 24 hours to allow reattachment. Reattached spheroid cells were stained with crystal violet and absorbance was quantified. Alternatively, cells in suspension were isolated and utilized for Cell TitreGLO cell viability measurements. *DYRK1A$^{-/-}$*spheroid cells had impaired survival compared to parental iOvCa147 spheroid cells at each time point and reattached and crystal violet stained cell viability matched Cell TitreGLO measurements of viability (one way anova, ****P<0.0001).

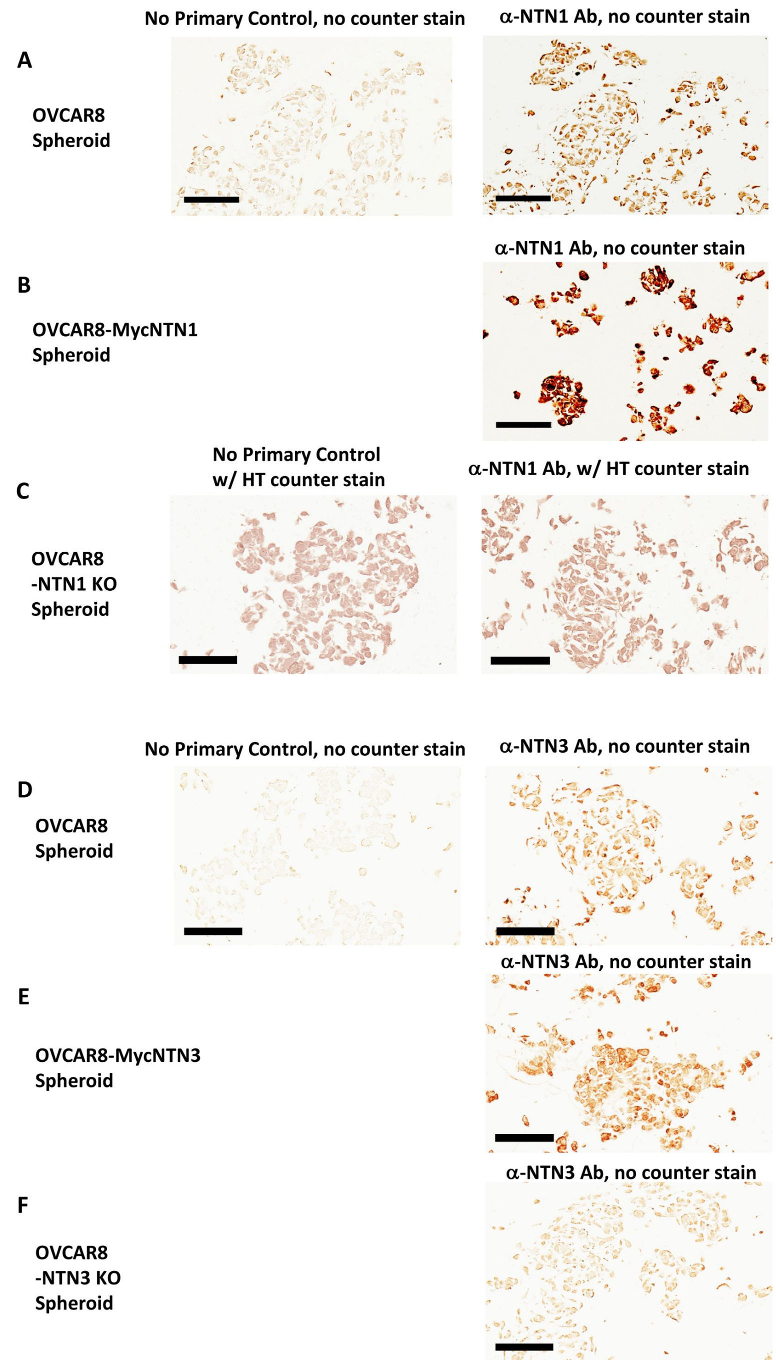

Figure 4 - Figure Supplement 1.

Specificity of Netrin-1 and Netrin-3 IHC staining of OVCAR8 spheroids.

A) OVCAR8 cells were allowed to form spheroids for six hours before fixation in formalin and embedding in paraffin for sectioning. Serial sections were stained with Netrin-1 Abs or a no primary antibody control. **B)** OVCAR8 cells overexpressing Netrin-1 were used to prepare spheroids as in A and stained in parallel to samples in A. **C)** OVCAR8 deleted for Netrin-1 expression were used to prepare spheroids as in A and serial sections were stained for Netrin-1 or with a no primary antibody control and counter stained. **D)** OVCAR8 cells were allowed to form spheroids for six hours before fixation in formalin and embedding in paraffin for sectioning. Serial sections were stained with Netrin-3 Abs or a no primary antibody control. **E)** OVCAR8 cells overexpressing Netrin-3 were used to prepare spheroids as in D and stained in parallel to samples in D. **F)** OVCAR8 deleted for Netrin-3 expression were used to prepare spheroids as in D and serial sections were stained for Netrin-3 or with a no primary antibody control. All scale bars = 100µm

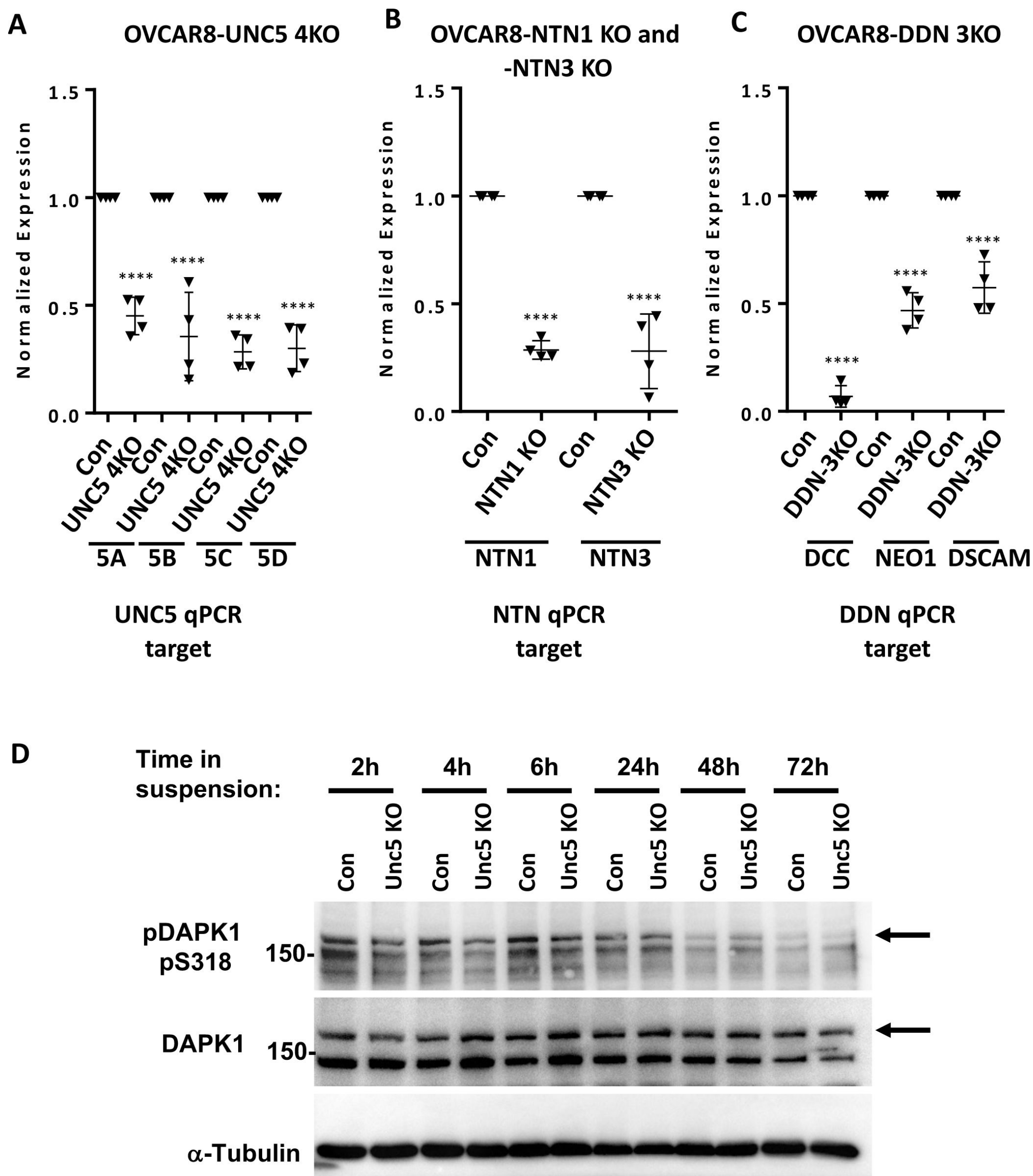

Figure 5 - Figure Supplement 1.

Evaluation of target gene transcript levels in multigene knock out cells.

A) RT-qPCR was performed on RNA isolated from control sgRNA expressing OVCAR8 cells and UNC5 4KO to assess expression of UNC5 family members. Graphs depict relative expression levels for each indicted gene in control and knock out cell lines. Mean expression levels of targeted cells were compared to background measurements using one way anova (****P<0.0001). **B)** RT-qPCR was used to measure mRNA levels of NTN1 and NTN3 transcripts in their respective CRISPR targeted, and OVCAR8 controls. Mean values were compared by one way anova (****P<0.0001). **C)** Levels of mRNA corresponding to DCC, DSCAM, and NEO1 were determined by RT-qPCR in DDN 3KO cells and control OVCAR8. Mean values were compared by one way anova (****P<0.0001). **D)** The indicated genotypes of cells were subjected to suspension culture and protein extracts were prepared at the indicated times. Proteins were resolved by SDS-PAGE, blotted, and probed with the indicated antibodies.

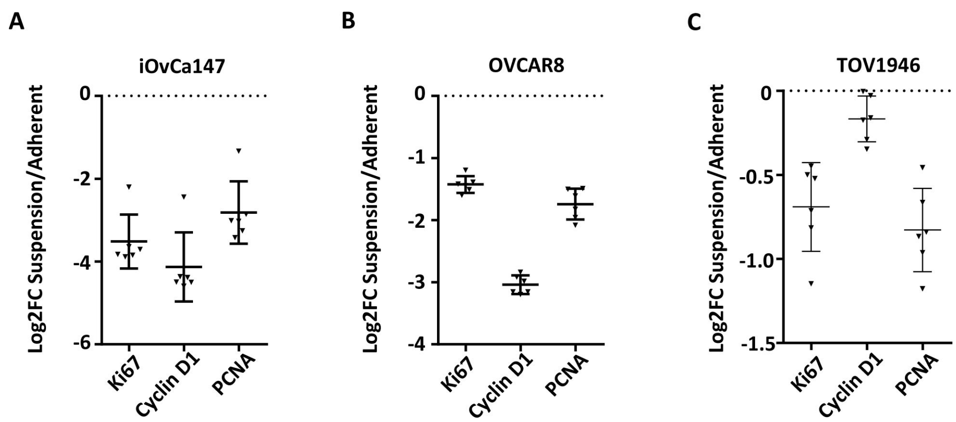

Figure 6 - Figure Supplement 1.

Evaluation of dormancy arrest in cell culture by RT-qPCR.

RNA was extracted from asynchronously proliferating cells and suspension cultures 72 hours post transfer from adherent. RT-qPCR was used to compare gene expression of the indicated cell cycle markers in iOvCa147, OVCAR8, TOV1946.

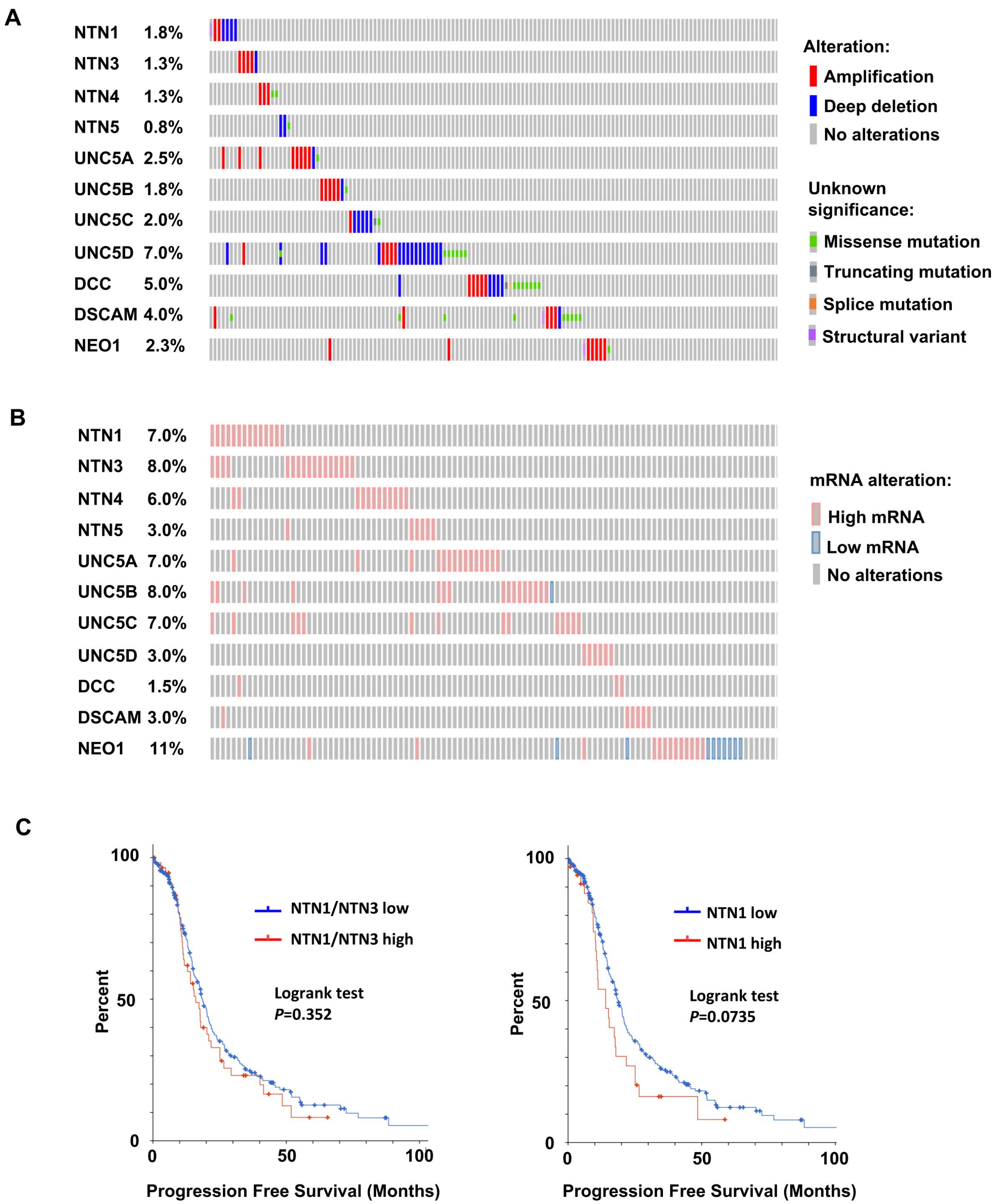

Figure 7 - Figure Supplement 1.

Frequency of Netrin ligand and receptor deletions, mutations or expression changes in high grade serous ovarian cancer.

A) Oncoprint of genetic alterations in 10 netrin signaling related genes in 398 patients. Alteration types are defined on the right. Only the first 154 patients are shown. **B)** Oncoprint illustrating gene expression outliers among 10 netrin signaling related genes in 201 patients (first 122 patients shown). Genomic data used is from Ovarian Serous Cystadenocarcinoma (TCGA, PanCancer Atlas) (201 samples with RNA seqV2, z-score cut off of 1.5). **C)** TCGA RNA-seq data for HGSOC patients (TCGA PanCancer Atlas study) was used to identify high Netrin-1 or -3 and low Netrin-1 or -3 expressing patients (high expressing are above z-score 1.2). Progression free survival was used to construct Kaplan-Meier plots and survival was compared using a logrank test.

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Editors

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Joint Public Review:

In this article, the authors employed modified CRISPR screens ["guide-only (GO)-CRISPR"] in the attempt to identify the genes which may mediate cancer cell dormancy in the high grade serous ovarian cancer (HGSOC) spheroid culture models. Using this approach, they observed that abrogation of several of the components of the netrin (e.g., DCC, UNC5Hs) and MAPK pathways compromise survival of non-proliferative ovarian cancer cells. This strategy was complemented by the RNAseq approach which revealed that number of the components of the netrin pathway are upregulated in non-proliferative ovarian cancer cells, and that their overexpression is lost upon disruption of DYRK1A kinase that has been previously demonstrated to play a major role in survival of these cells. Perampalam et al. then employed a battery of cell biology approaches to support the model whereby the Netrin signaling governs the MEK-ERK axis to support survival of non-proliferative ovarian cancer cells. Moreover, the authors show that overexpression of Netrins 1 and 3 bolsters dissemination of ovarian cancer cells in the xenograft mouse model, while also providing evidence that high levels of the aforementioned factors are associated with poor prognosis of HGSOC patients.

Strengths:

In this valuable study Perampalam et al. developed a CRISPR-based screening approach to identify key genes that are enriched in high grade serous ovarian cancer spheroids. This led to a discovery that Netrin signaling plays a prominent role in survival of ovarian cancer cells.

During revision, the authors provide additional evidence to support their central claims and to this end, it was found that they now provide solid evidence to substantiate the proposed model. This work is anticipated to be of interest to cancer biologists specializing in ovarian cancer biology.

https://doi.org/10.7554/eLife.91766.2.sa0

Author response:

The following is the authors' response to the original reviews.

> ***Reviewer #1 (Public Review):***
>
> *Summary:*
>
> *Perampalam et al. describe novel methods for genome-wide CRISPR screening to identify and validate genes essential for HGSOC spheroid viability. In this study, they report that Netrin signaling is essential for maintaining disseminated cancer spheroid survival, wherein overexpression of Netrin pathway genes increases tumor burden in a xenograft model of ovarian cancer. They also show that high netrin expression correlates with poor survival outcomes in ovarian cancer patients. The study provides insights into the biology of netrin signaling in DTC cluster survival and warrants development of therapies to block netrin signaling for treating serous ovarian cancer.*
>
> *Strengths:*
>
> *- The study identifies Netrin signaling to be important in disseminated cancer spheroid survival*
>
> *- A Novel GO-CRISPR methodology was used to find key genes and pathways essential for disseminated cancer cell survival*

Thanks for the endorsement of our work and its importance to metastasis in ovarian cancer.

> *Weaknesses:*
>
> *- The term dormancy is not fully validated and requires additional confirmation to claim the importance of Netrin signaling in "dormant" cancer survival.*
>
> *- Findings shown in the study largely relate to cancer dissemination and DTS survival rather than cancer dormancy.*

Much of the validation of dormancy and cell cycle arrest in HGSOC spheroids, as well as the culture model, have been published previously and hence was not repeated here. I think this reviewer will appreciate the updated citations and explanations to better illustrate the state of knowledge. We have also added new experiments that further emphasize the dormant state of spheroid cells in culture and xenografts, as well as patient derived spheroids used in this study.

> ***Reviewer #1 (Recommendations for Authors):***
>
> *(1) It is unclear what spheroid/adherent enrichment ratio is and how it ties into genes affecting cell viability. Why is an ER below 1 the criteria for selecting survival genes?*

Our screen uses the 'guide only' comparison in each culture condition to establish a gene score under that specific condition. A low adherent score captures genes that are essential under standard culture conditions where cells are proliferating and this can include genes needed for proliferation or other basic functions in cell physiology. A low spheroid score identifies the genes that are most depleted in suspension when cells are growth arrested and this is an indication of cell death in this condition. Since gene knock outs are first established in adherent proliferating conditions, essential genes under these conditions will already start to become depleted from the population before suspension culture. By selecting genes with a ratio of <1 we can identify those that are most relevant to dormant suspension culture conditions. Ultimately, the lowest enrichment ratio scores represent genes whose loss of function is dispensable in the initial adherent condition, but critical for survival in suspension and this is what we aimed to identify. We've updated Figure 1B to illustrate this and we've updated the explanation of the enrichment ratio on page 6, lines 144 to 147 of the results.

> *(2) The WB for phospho-p38 in figure 1A for OVCAR8 line does not show increased phosphorylation in the spheroid relative to the adherent. If anything, phospho-p38 appears to be reduced in the spheroid. Can the authors provide a better western blot?*

We've updated this blot with a longer exposure, see Figure 1A. Phosphorylation levels of p38 are essentially unchanged in OVCAR8 cells in suspension culture, although the overall levels of p38 may be slightly reduced in dormant culture conditions.

> *(3) How did the authors confirm dormancy apart from western blot for phospho-ERK vs phospho-p38? Authors should add EdU/BrdU staining and/or Ki67 staining to confirm dormancy.*

Previous publications that appear as citations 7,10, and 33 in the reference list established the growth arrest state of these cells in suspension culture in the past. This included measuring other known markers of dormancy and quiescence such as p27, p130, and reduced cyclin/cdk activity and 3H-thymidine incorporation. In addition, other associated characteristics of dormancy such as EMT and catabolic metabolism have been demonstrated in these culture conditions (see citation 11 and Rafehi et al. Endocr. Relat. Cancer 23;147-59). We've added these additional citations to our descriptions of dormant spheroid culture to better clarify the status of these cells in our experiments (see page 6, lines 126-28). To ensure that cells are growth arrested in the experiments shown in this paper, we have updated Figure 1A to include blots of p130 and Ki67 to further emphasize that spheroid cells are not proliferating as the quiescence marker (p130) is high and the proliferative marker (Ki67) is lost in suspension culture.

> *(4) Can the authors report spheroid volume over time in culture? How was viability measured?*

We've updated the methods (see page 27, line 574) to better highlight the description of cell survival that answers both of these questions. At the ends of experimental time points in both the screen and viability assays we captured live cells by replating on adherent plasticware. We fixed and stained with crystal violet and photographed plates to illustrate the sizes of spheroids (shown in Fig. 2 Supplement 1E, Fig. 6C, and 7D). We subsequently extracted the dye and quantitated it spectrophotometrically to quantitatively compare biomass of viable cells between experiments irrespective of the relatively random shapes of spheroids. We found reattachment and staining in this manner to match traditional viability assays such as CellTiter-Glo in a previous paper (10). Furthermore, biomass never increases in culture and diminishes gradually over time in culture consistent with the non-proliferative state of these experiments. Double checks of this equivalency of viability and reattached biomass measurments, as well as demonstrating that biomass is lost over time, are shown in Fig. 2 Supplement 1E that compares reattached crystal violet staining measurements with CellTiter-

Glo for DYRK1A knock out cells over time in culture. In addition, we include a comparison of crystal violet staining of reattached spheroids with trypan blue dye exclusion in Fig. 5G and H. In both cases reattachment and more direct viability assays demonstrate the same conclusion that Netrin signaling supports viability in dormant culture.

> *(5) Please show survival significance of Netrin signaling genes in recurrence/relapse free survival to claim importance in cancer dormancy.*

See Fig. 7 Supplement 1C where we include the recurrence free survival data. Netrin-1, and -3 high expressors also have a numerically shorter progression free survival but it is not statistically significant. Netrin-1 overexpression alone is also shown and it shows shorter survival with a P-value of 0.0735. Elevated survival of dormant cells in a residual disease state is expected to increase the chance of relapse and shorten this interval. Thus, this data is consistent with our model, but lacks statistical significance.

There are many alternative ways to interpret what shorter progression free survival, or overall survival, may mean biologically. Since survival of dormant cells is but one of them, we also added new data to experimentally investigate the role of endogenous Netrin signaling in dormant residual disease in Fig. 6 and described on page 12, lines 266-87. We used xenograft experiments to show OVCAR8 spheroids form and withdraw from the cell cycle equivalently to suspension culture following intraperitoneal injection. Furthermore, loss of Netrin signaling due to receptor deletions compromises survival during this early window before disseminated lesions form. This argues that Netrin signaling contributes to survival during this window of dormancy. In addition, mice engrafted with mutant cells experience prolonged survival when Netrin signaling is blocked. Together, these experiments further argue that Netrin signaling supports survival in the dormant, non-proliferative phase, and leads to reduced survival of mice.

> *(6) The authors show IHC staining of patient ascites derived HGSOC spheroids. However, no marker for dormancy is shown in these spheroids. Adding Ki67 staining or phospho-ERK vs phospho-p38 would be necessary to confirm cancer dormancy.*

We have added new staining for Ki67 and p130 that compares these markers in HGSOC tumors where Ki67 is high and p130 is low with ascites derived spheroids where staining is the opposite. Importantly, expression of p130 is linked to cellular quiescence and is not found to accumulate in the nucleus of cells that are just transiting through G1. This confirms that the ascites derived spheroids are dormant. See Fig. 4A-E and described on page 9, lines 201-7.

> *(7) Overall, the findings are interesting in the context of cancer dissemination. There is not enough evidence for cancer dormancy and the importance of Netrin signaling in the survival of cancer dormancy. Overexpression of Netrin increases phosphorylation of ERK, leading one to expect an increase in proliferation. This suggests that Netrin breaks cancer cells out of dormancy, into a proliferative state.*

We have found that the discovery of Netrin activation of MEK-ERK in growth arrested cells is counterintuitive to many cancer researchers. However, this axis exists in other paradigms of Netrin signaling in axon outgrowth that are not proliferation related (see citation 26, Forcet et al. Nature 417; 443-7 as an example). We have added Fig. 5D and descriptions on page 11, lines 244-52 to better clarify that Netrins CAN'T induce cell proliferation through ERK. Addition of recombinant Netrin-1 can only induce ERK phosphorylation in suspension culture conditions and not in quiescent adherent conditions. The small magnitude of ERK phosphorylation induced by Netrin-1 in suspension compared to treating adherent, quiescent cells with the same concentration of mitogenic EGF further emphasizes that this is not a proliferative signal. Lastly, the new xenograft experiment in Fig. 6A-D (described on page 12, lines 266-81 demonstrates the growth arrested context in which Netrin signaling in dormant spheroids leads supports viability.

> *(8) If authors wish to claim cancer dormancy as the premise of their study, additional confirmatory experiments are required to support their claims. Alternatively, based on the current findings of the study, it would be best to change the premise of the article to Netrin signaling in cancer dissemination and survival of disseminated cancer spheroids rather than cancer dormancy.*

I expect that this reviewer will agree that we have added more than sufficient explanations of background work on HGSOC spheroid dormancy from the literature, as well as new experiments that address their questions about dormancy in our experiments.

> ***Reviewer #2 (Public Review):***
>
> ***Summary:***
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> *In this article, the authors employed modified CRISPR screens ["guide-only (GO)-CRISPR"] in the attempt to identify the genes which may mediate cancer cell dormancy in the high grade serous ovarian cancer (HGSOC) spheroid culture models. Using this approach, they observed that abrogation of several of the components of the netrin (e.g., DCC, UNC5Hs) and MAPK pathways compromise the survival of non-proliferative ovarian cancer cells. This strategy was complemented by the RNAseq approach which revealed that a number of the components of the netrin pathway are upregulated in non-proliferative ovarian cancer cells and that their overexpression is lost upon disruption of DYRK1A kinase that has been previously demonstrated to play a major role in survival of these cells. Perampalam et al. then employed a battery of cell biology approaches to support the model whereby the Netrin signaling governs the MEK-ERK axis to support survival of non-proliferative ovarian cancer cells. Moreover, the authors show that overexpression of Netrins 1 and 3 bolsters dissemination of ovarian cancer cells in the xenograft mouse model, while also providing evidence that high levels of the aforementioned factors are associated with poor prognosis of HGSOC patients.*
>
> *Strengths:*
>
> *Overall it was thought that this study is of potentially broad interest in as much as it provides previously unappreciated insights into the potential molecular underpinnings of cancer cell dormancy, which has been associated with therapy resistance, disease dissemination, and relapse as well as poor prognosis. Notwithstanding the potential limitations of cellular models in mimicking cancer cell dormancy, it was thought that the authors provided sufficient support for their model that netrin signaling drives survival of non-proliferating ovarian cancer cells and their dissemination. Collectively, it was thought that these findings hold a promise to significantly contribute to the understanding of the molecular mechanisms of cancer cell dormancy and in the long term may provide a molecular basis to address this emerging major issue in the clinical practice.*

Thanks for the kind words about the importance of our work in the broader challenges of cancer treatment.

> *Weaknesses:*
>
> *Several issues were observed regarding methodology and data interpretation. The major concerns were related to the reliability of modelling cancer cell dormancy. To this end, it was relatively hard to appreciate how the employed spheroid model allows to distinguish between dormant and e.g., quiescent or even senescent cells. This was in contrast to solid*

evidence that netrin signaling stimulates abdominal dissemination of ovarian cancer cells in the mouse xenograft and their survival in organoid culture. Moreover, the role of ERK in mediating the effects of netrin signaling in the context of the survival of non-proliferative ovarian cancer cells was found to be somewhat underdeveloped.

Experiments previously published in citation 7 show that growth arrest in patient ascites derived spheroids is fully reversible and that argued against non-proliferative spheroids being a form of senescence and moved this work into the dormancy field. We have added extensive new support for our model systems and data to address the counterintuitive aspects of MEK-ERK signaling in survival instead of proliferation.

Reviewer #1 Recommendations for Authors

(1) A better characterization of the spheroid model may be warranted, including staining for the markers of quiescence and senescence (including combining these markers with staining for the components of the netrin pathway)

See Figure 1A and page 6, lines 126-36 where we have added blots for Ki67 and p130 to better emphasize the arrested proliferative state of cells in our screening conditions. We have also added these same controls for patient ascites-derived spheroids in Figure 4 and described on page 9, lines 203-7. One realization from this CRISPR screen, and others in our lab, is that it identifies functionally important aspects of cell physiology and not necessarily ones that are easily explored using commercially available antibodies. Netrin-1 and -3 staining of patient derived spheroids in Fig. 4, as well as cell line spheroids stained in Fig. 4 Supplement 1 further support the relevance of this pathway in dormant cancer cells because Netrins are expressed in the right place at the right time. The Netrin-1 stimulation experiments in Fig. 5C were originally carried out to probe HGSOC cells for functionality of Netrin receptors since we couldn't reliably detected them by blotting or staining with available antibodies. This demonstrates that this pathway is active in the various HGSOC cell lines we've used and specifically, using OVCAR8 cells, we show it is only active in suspension culture conditions.

(2) In figure 1A it appears that total p38 levels are reduced in some cell lines in spheroid vs. adherent culture. The authors should comment on this.

These blots have been updated to be more clear. Overall p38 levels may be reduced in some cell lines and when compared with activation levels of phosphorylated p38 it suggests the fraction of activated p38 is higher. OVCAR8 cells may be an exception where the overall activity level remains approximately the same.

(3) The authors should perhaps provide a clearer rationale for choosing to focus on the netrin signaling vs. e.g., GPCR signaling, and consider more explicit defining of "primary" vs. "tertiary" categories in Reactome gene set analysis.

We've updated Fig. 1E and the text on page7, lines 161-5 to illustrate which gene categories identified in the screen belong to which tiers of Reactome categories. It better visualizes why we have investigated the Axon guidance pathway that includes Netrin because it is a highly specific signaling pathway that scores similarly to the broader and less specific categories at the very top of the list. As an aside, the GPCR signaling and GPCR downstream signaling have proven to be fairly intractable categories. As best we can tell the GPCR downstream signaling category is full of MAPK family members and likely represents some redundancy with MAPK further down.

(4) In figure 3A-C, including factors whose expression did not appear to change between adherent and suspension conditions may be warranted as the internal control. Figure

> *3D-F may benefit from some sort of quantification.*

The mRNA expression levels are normalized to GAPDH as an internal control. We have updated this figure and re-plotted it as fold change relative to adherent culture cells with statistical comparisons to indicate which are significantly upregulated in suspension culture.

The IHC experiments are now in Fig. 4D-F and show positive staining for Netrin-1 and -3. Netrin-3 is easiest to see, while Netrin-1 is trickier because the difference with the no primary antibody control isn't intensity, but the tint of the DAB stain. We had to counter stain the patient spheroids with Hematoxylin in order for the slide scanner to find the best focal plane and make image registration between sections possible. This unfortunately makes the Netrin-1 staining rather subtle. For cell line spheroids in the Fig. 4, Supplement 1 we didn't need the slide scanner and show negative controls without counter stain that are much more convincing of Netrin-1 detection and reassure us that our staining detects the intended target. We've updated the labels in Fig. 4 and Fig. 4, Supplement 1 for this to be more intuitive. Unfortunately, relying on the tint of the DAB stain leaves this as a qualitative experiment.

> *- In figure 4C-E the authors show that Netrin-1 stimulation induces ERK phosphorylation whereby it is argued that this is a "low-level" stimulation of ERK signaling required for the survival of ovarian cells in the suspension. This is however hard to appreciate, and it was thought that having adherent cells in parallel would be helpful to wage whether this indeed is a "low level" ERK activity. Moreover, the authors should likely include downstream substrates of ERK (e.g., RSKs) as well as p38 in these experiments. The control experiments for the effects of PD184352 on ERK phosphorylation also appear to be warranted. Finally, performing the experiments with PD184352 in the presence of Netrin-1 stimulation would also be advantageous.*

We have added a new Netrin-1 stimulation experiment in Fig. 4D (described on page 11, line 244-52) that shows that Netrins can only activate very low levels of ERK phosphorylation in suspension when proliferation is arrested. Netrin-1 stimulation of quiescent adherent cells where stimulation of proliferation is possible shows that Netrins are unable to activate ERK phosphorylation in this condition. In contrast, we also stimulate quiescent adherent OVCAR8 cells with an equal concentration of EGF (a known mitogen) to offer high level ERK phosphorylation as a side by side comparison. I think that this offers clear evidence that Netrin signaling is inconsistent with inducing cell proliferation. We've also updated citations in the introduction to include citation 26 that offers a previously reported paradigm of Netrin-ERK signaling in axon outgrowth that is a non-cancer, non-proliferative context to remind readers that Netrins utilize MEK-ERK differently.

We highlight Netrin-MEK-ERK signaling as key to survival for a number of reasons. First, Netrin signaling in this paradigm does not fit the dependence receptor paradigm where loss of Netrin receptors protect against cell death. Fig. 5B rules this out as receptor loss never offers a survival advantage, but clearly receptor deletions compromise survival in suspension culture. Second, positive Netrin signaling is known to support survival by inactivating phosphorylation of DAPK1. We've added this experiment as Fig. 5 Supplement 1D and show that loss of Netrin receptors doesn't reduce DAPK1 phosphorylation in a time course of suspension culture. Consequently, we conclude this isn't the survival signal either. Since MEK and ERK family members scored in our screen, we investigated their role in survival. We now show two different MEK inhibitors with different inhibitory mechanisms to confirm that MEK inhibition induces cell death. In addition to the previous PD184352 inhibitor in our first submission, we've added Trametinib as well and this is shown in Fig. 5G. Since it is surprising the MEK inhibition can kill instead of just arrest proliferation, we've also added another cell death assay in which we show trypan blue dye exclusion as a second look at survival. This is now Fig. 5H. Lastly, we include Trametinib inhibition of ERK phosphorylation in these assays in Fig. 5I. While we leave open what takes place downstream of ERK, our model in Fig. 5J offers a very detailed look at the components upstream.

> *- Does inhibition of ERK prevent the abdominal spread of ovarian cancer cells? The authors may feel that this is out of the scope of the study, which I would agree with, but then the claims regarding ERK being the major mediator of the effects of netrin signaling should be perhaps slightly toned down.*

We agree that loss of function xenograft experiments will enhance our discovery of Netrin's role in dormancy and metastasis. We have added a new Fig. 6 that uses xenografts with Netrin receptor deficient OVCAR8 cells (UNC5 4KO). It demonstrates that two weeks following IP engraftment we can isolate spheroids from abdominal washes and that cells have entered a state of reduced proliferation as determined by lowered Ki67 expression as well as other proliferation inducing genes. In the case of UNC5 4KO cells, there is significant attrition of these cells as determined by recovering spheroids in adherent culture (Fig.6C) and by Alu PCR to detect human cells in abdominal washes (Fig. 6D). Lastly, xenografts of UNC5 4KO cells cause much less aggressive disease and significantly extend survival of these mice (Fig. 6E,F). Not exactly the experiment that the reviewer is asking for, but a clear indication that Netrin signaling supports survival in xenograft model of dormancy.

> *- Notwithstanding that this could be deduced from figures 6D and F, it would be helpful if the number of mice used in each experimental group is clearly annotated in the corresponding figure legends. Moreover, indicating the precise statistical tests that were used in the figures would be helpful (e.g., specifying whether anova is one-way, two-way, or?)*

We have added labels to what is now Fig. 8B to indicate the number of animals used for each genotype of cells. We have also updated figure legends to include more details of statistical tests used in each instance.

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